MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: DESIGN CALCULATION 6000 VILLAGE:CLASS-C ACCOMMODATION DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
Typ.	Typical
HCS	Hollow Core Slab
Fc'	Concrete Compressive Strength
Fy	Yield Strength of Reinforcing Steel
ϕ	Strength Reduction Factor
SBC	Saudi Building Code
SASO	Saudi Standards, Metrology and Quality Organization
ACI	American Concrete Institute
ASCE	American Society of Civil Engineers
PCI	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of an Accommodation C Building.

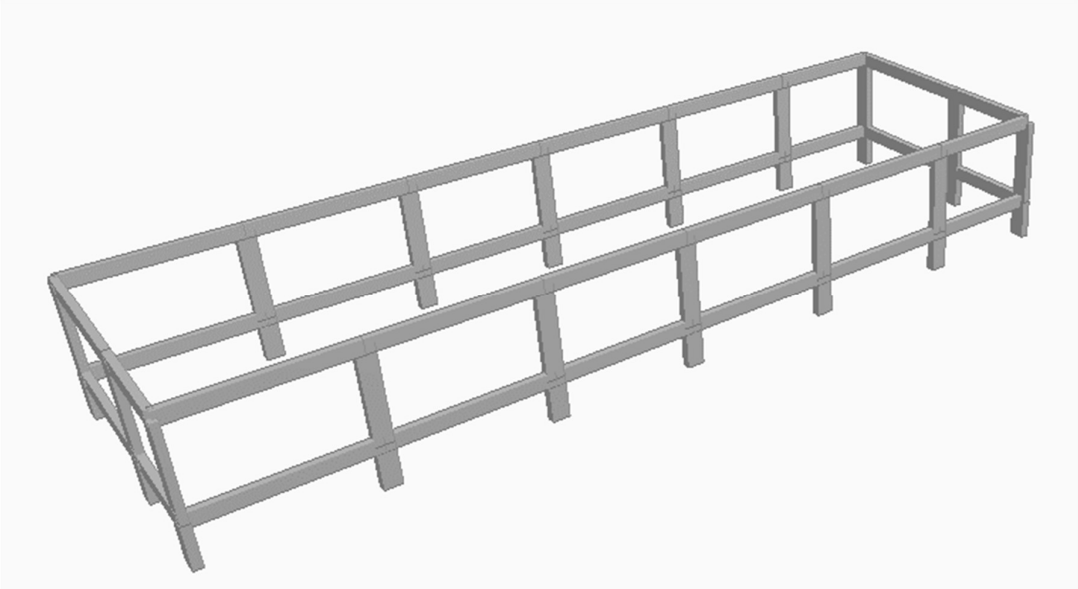
2. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Accommodation C Building only.

3. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HSC (Hollow Core Slab)

The Superstructure for this Building was modelled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

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3.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

3.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

3.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

4. DESIGN LOADS

4.1. Dead Load

4.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab

(HCS-300 Topped)(To be confirmed by Vendor) = **3.88 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load =3.88+0.2+3.1= 7.2 **kN/m²**

UDL on Edge Ledger Beams (LB1) = 7.2 x 9.67/2 = **34.82 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+(0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

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4.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

4.2. Live Load

4.2.1. Roof Live Loads

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

UDL Due to Live Load on the Edge Ledger Beam (LB1& LB2) =0.5X9.67/2= **2.42 kN/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

4.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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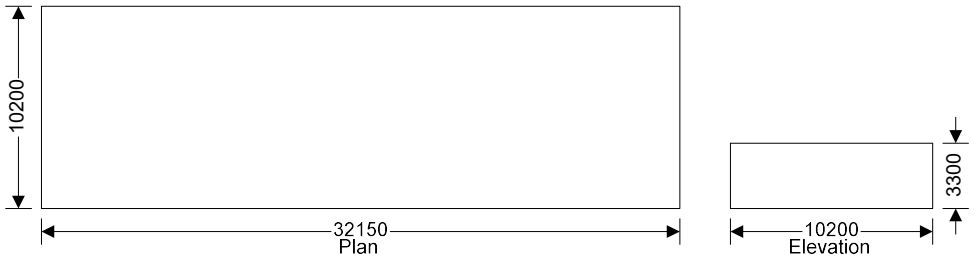
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4.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = **32150** mm
Width of building;	d = **10200** mm
Height to eaves;	H = **3300** mm
Mean height;	h = **3300** mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant;	K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

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Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m²)
3300	1.03	1.29
3301	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m²)	Net pressure p (kN/m²)	Area A_{ref} (m²)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	106.13	17.78
B	3.30	-0.50	1.29	-1.26	106.10	-133.29
C	3.30	-0.70	1.29	-1.48	33.66	-49.66
D	3.30	-0.70	1.29	-1.48	33.66	-49.66

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H =$ **106.10** m²

Projected vertical area of roof; $A_{vert_r_0} =$ **0.00** m²

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} =$ **81.7** kN

Leeward net force; $F_l = F_{w,wB} =$ **-133.3** kN

Windward net force; $F_w = F_{w,wA} =$ **17.8** kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) =$ **151.1** kN

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	106.13	168.20
B	3.30	-0.50	1.29	0.16	106.10	17.09
C	3.30	-0.70	1.29	-0.06	33.66	-1.95
D	3.30	-0.70	1.29	-0.06	33.66	-1.95

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H =$ **106.10** m^2

Projected vertical area of roof; $A_{vert_r_0} =$ **0.00** m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} =$ **81.7** kN

Leeward net force; $F_l = F_{w,wB} =$ **17.1** kN

Windward net force; $F_w = F_{w,wA} =$ **168.2** kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) =$ **151.1** kN

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	33.67	5.64
B	3.30	-0.24	1.29	-0.97	33.66	-32.79
C	3.30	-0.70	1.29	-1.48	106.10	-156.53
D	3.30	-0.70	1.29	-1.48	106.10	-156.53

Overall loading

Projected vertical plan area of wall; $A_{vert_w_90} = d \times H =$ **33.66** m^2

Projected vertical area of roof; $A_{vert_r_90} =$ **0.00** m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} =$ **25.9** kN

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Leeward net force; $F_l = F_{w,wB}$ = **-32.8** kN

Windward net force; $F_w = F_{w,wA}$ = **5.6** kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min})$ = **38.4** kN

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m²)	Net pressure p (kN/m²)	Area A_{ref} (m²)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	33.67	53.37
B	3.30	-0.24	1.29	0.44	33.66	14.92
C	3.30	-0.70	1.29	-0.06	106.10	-6.15
D	3.30	-0.70	1.29	-0.06	106.10	-6.15

Overall loading

Projected vertical plan area of wall; $A_{vert_w_90} = d \times H$ = **33.66** m²

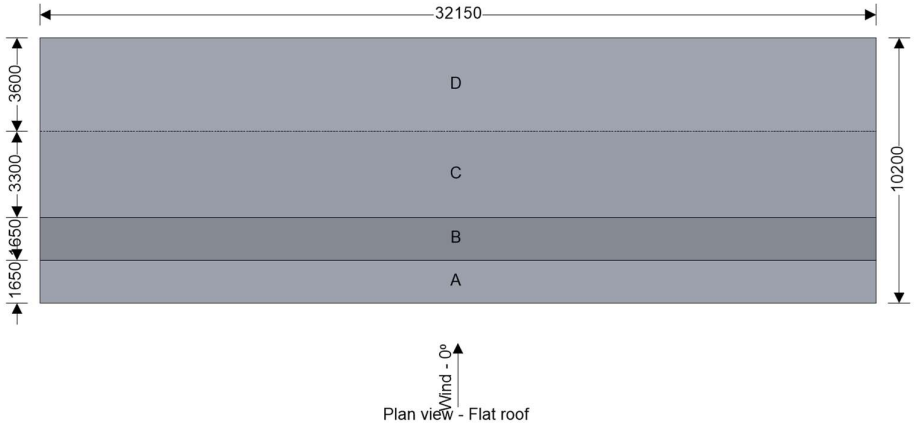
Projected vertical area of roof; $A_{vert_r_90}$ = **0.00** m²

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90}$ = **25.9** kN

Leeward net force; $F_l = F_{w,wB}$ = **14.9** kN

Windward net force; $F_w = F_{w,wA}$ = **53.4** kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min})$ = **38.4** kN

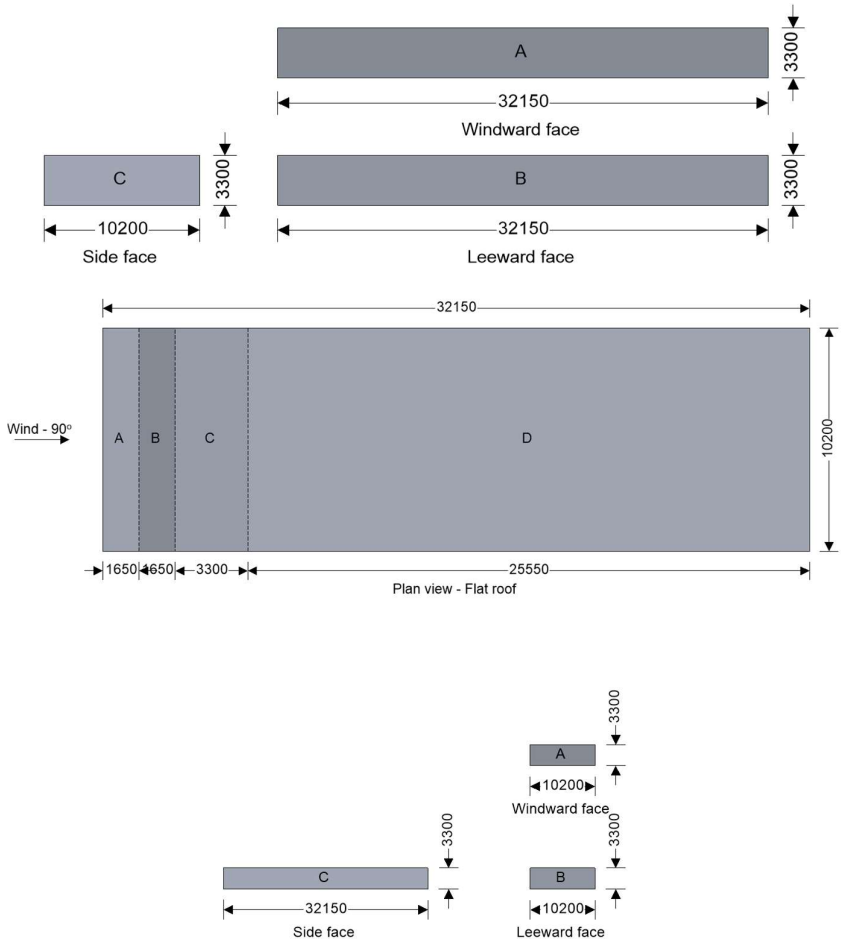


Plan view - Flat roof

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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4.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

4.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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5. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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6. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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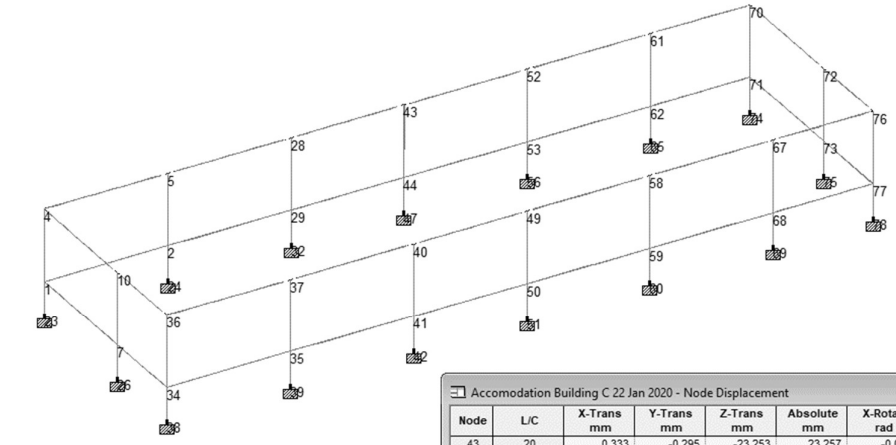
7. DEFLECTIONS/DRIFT

7.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Accomodation Building C 22 Jan 2020 - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
43	20	0.333	-0.295	-23.253	23.257	-0.008	0.000	-0.000
43	16	0.309	-0.177	-22.352	22.355	-0.007	0.000	-0.000
52	20	0.313	-0.311	-21.547	21.551	-0.007	-0.001	-0.000
28	20	0.352	-0.289	-21.301	21.306	-0.007	0.000	0.000
52	16	0.298	-0.187	-20.758	20.761	-0.007	-0.001	-0.000
28	16	0.322	-0.173	-20.434	20.438	-0.007	0.000	-0.000
49	21	0.248	-0.295	19.717	19.721	0.007	0.000	-0.000
43	17	0.271	-0.177	19.714	19.717	0.006	-0.000	-0.000
49	20	0.236	-0.295	19.699	19.703	0.007	0.000	-0.000
40	21	0.258	-0.289	18.872	18.876	0.006	-0.000	0.000
49	17	0.258	-0.177	18.823	18.826	0.006	0.000	-0.000
40	20	0.248	-0.289	18.817	18.821	0.006	-0.000	0.000
43	21	0.295	-0.295	18.814	18.819	0.006	-0.000	-0.000

Maximum attained Drift as per Analysis from above figure = 23mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3300/100 = 33mm

Hence the Structure is safe considering drift check…

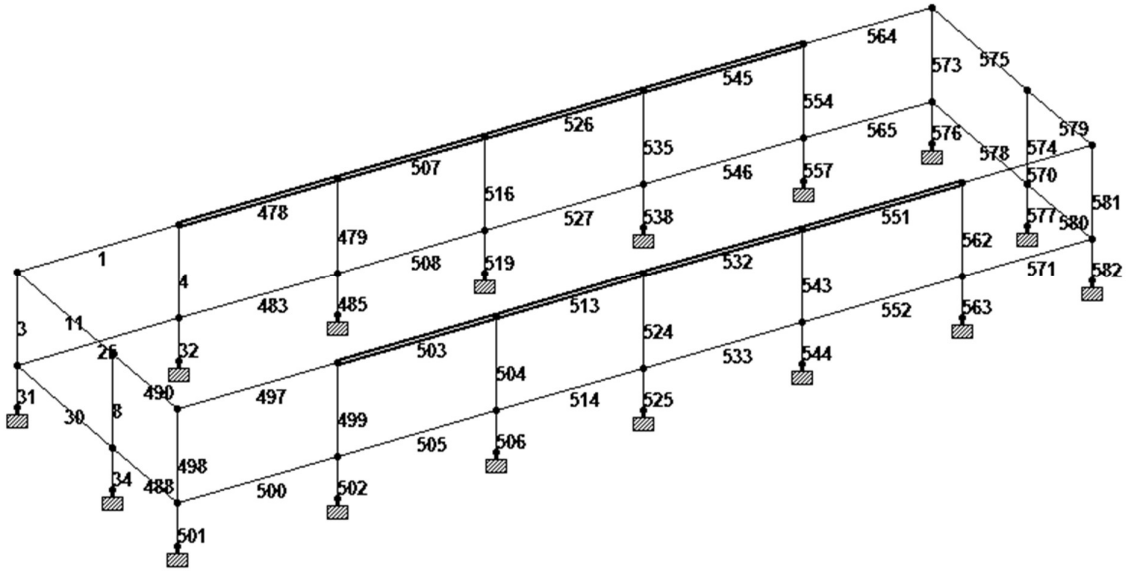
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8. CONCRETE ROOF BEAM DESIGN

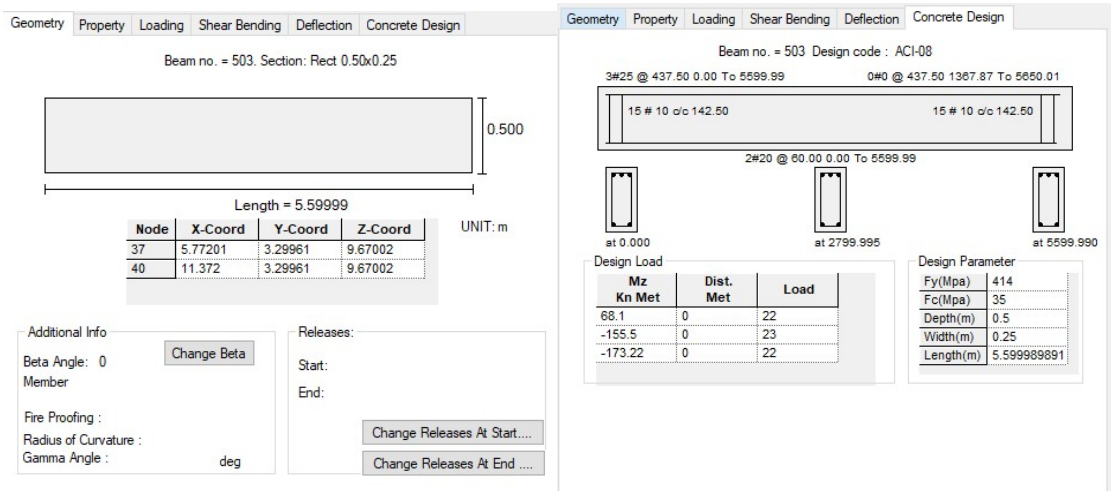
8.1. Design of Ledger Beam(LB1)



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Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 173 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1137 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **68 Knm (Sagging)**

Ast2.req = **426 mm²**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **1470mm² >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm² >>>>>>(Ast.min, Ast2.req)**

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8.1.1. Staad Output for Lb1

BEAM NO. 503 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

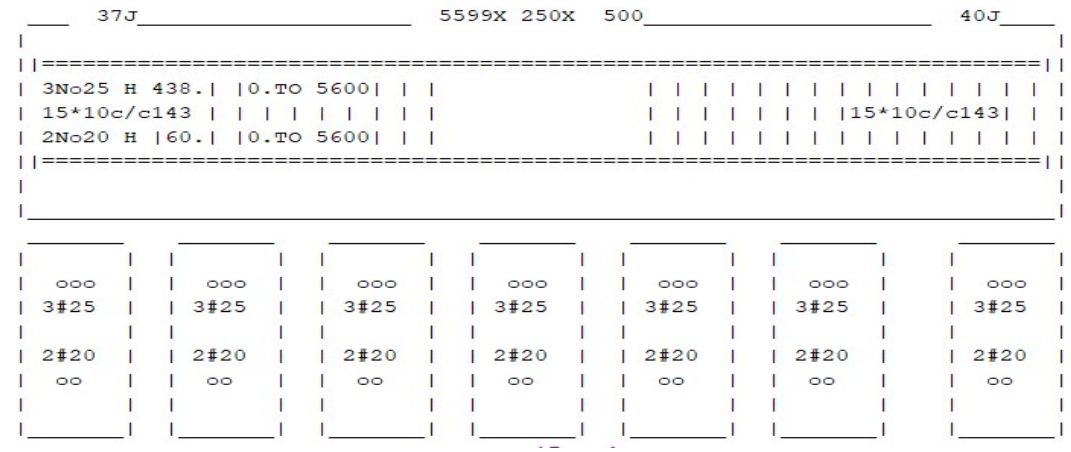
LEN - 5600. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	0.	5600.	YES	YES
2	438.	3 - 25MM	0.	5600.	YES	YES

B E A M N O. 503 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 94.49 KNS Vc= 113.79 KNS Vs= 12.19 KNS
Tu= 11.70 KN-MET Tc= 4.0 KN-MET Ts= 15.6 KN-MET LOAD 27
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 1920. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.86 SQ.CM.

AT END SUPPORT - Vu= 64.21 KNS Vc= 118.45 KNS Vs= 0.00 KNS
Tu= 7.15 KN-MET Tc= 4.0 KN-MET Ts= 9.5 KN-MET LOAD 31
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.

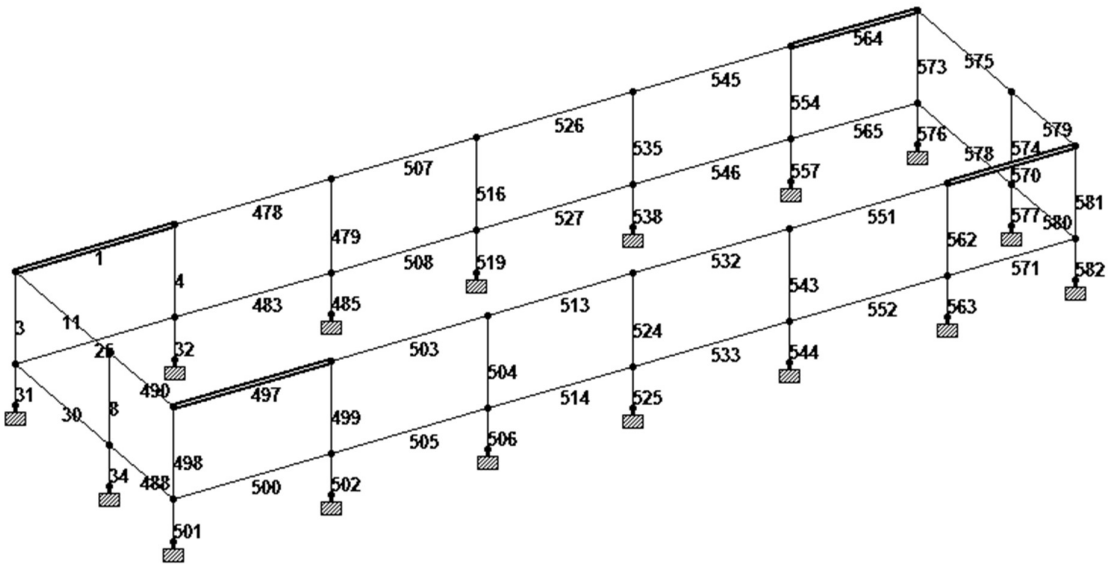


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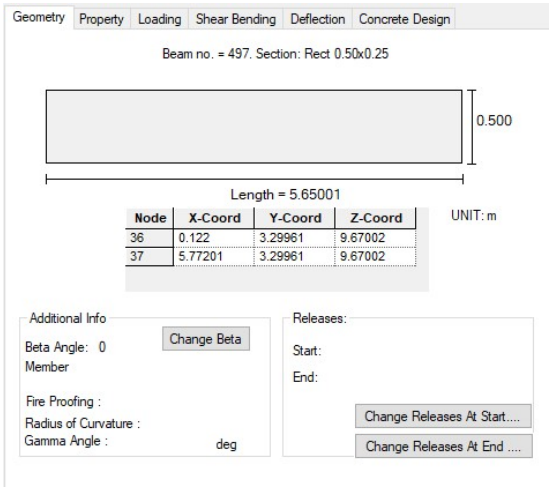
8.2. Design of Ledger Beam(LB2)

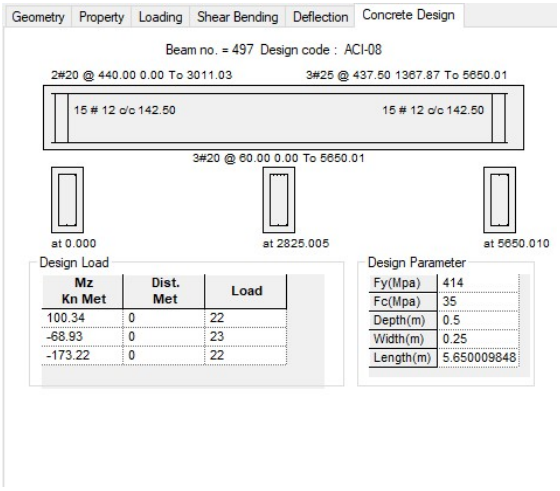


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Design of Ledger Beam (LB2)

Maximum Moment required from Analysis = 173 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1137 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **100 Knm (Sagging)**

Ast2.req = **635 mm²**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **1470mm² >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm² >>>>>>(Ast.min, Ast2.req)**

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8.2.1. Staad Output For Lb2

```
        BEAM  NO.   497 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

 LEN -   5650. MM  FY -   414.  FC -   35.  MPA, SIZE -   250. X  500. MMS

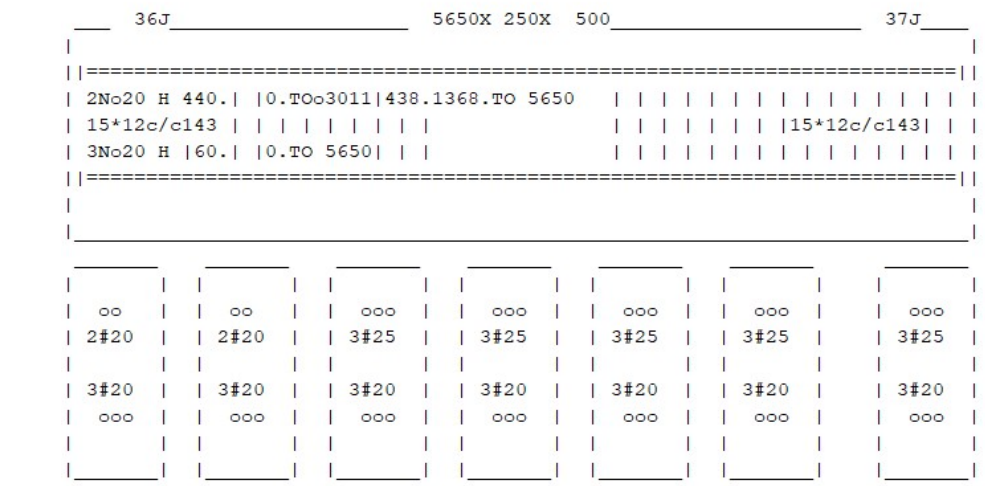
 LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
             (MM)                        (MM)          (MM)       STA  END
 ____________________________________________________________________________


   1          60.      3 - 20MM          0.          5650.        YES  YES
   2         440.      2 - 20MM          0.          3011.        YES  NO
   3         438.      3 - 25MM       1368.          5650.        NO   YES


       B E A M  N O.   497 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    72.56 KNS  Vc=   114.27 KNS  Vs=     0.00 KNS
 Tu=    22.65 KN-MET  Tc=     4.1 KN-MET  Ts=    30.2 KN-MET  LOAD     27
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1945. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    7.46 SQ.CM.

 AT END   SUPPORT - Vu=   114.02 KNS  Vc=   117.91 KNS  Vs=    34.11 KNS
 Tu=    17.59 KN-MET  Tc=     4.1 KN-MET  Ts=    23.5 KN-MET  LOAD     27
 STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
```

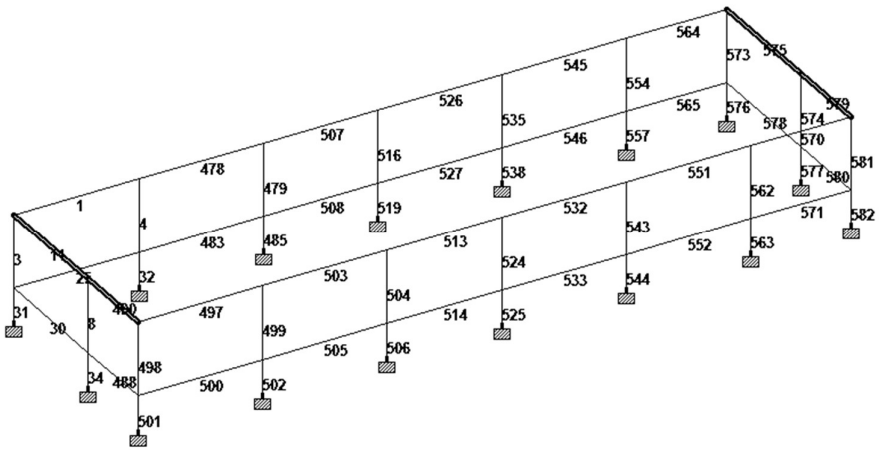


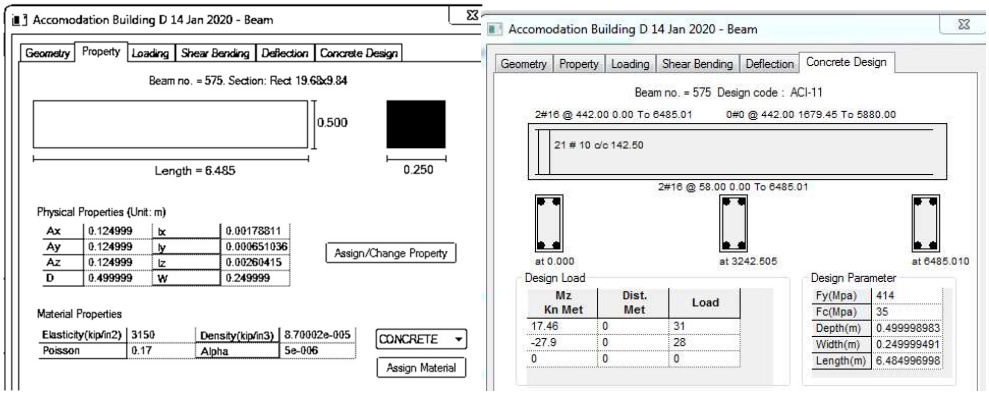
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8.3. Design of Beam-(B1)





Ast.min= **406mm²**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

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8.3.1. Staad Output For B1

```
ACI 318-11   BEAM  NO.   575   DESIGN RESULTS
==============================================

LEN -  6485. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  500. MMS

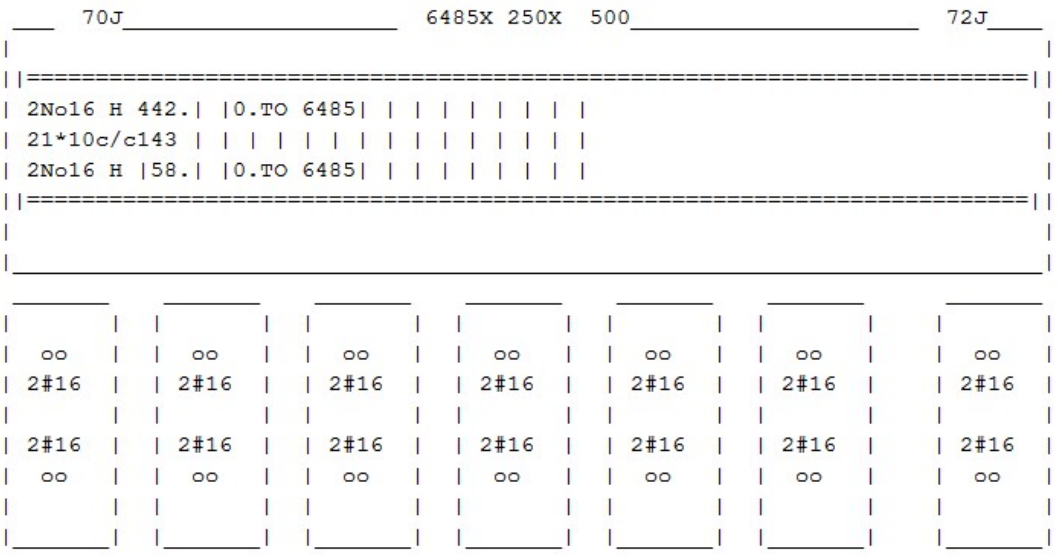
LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                        (MM)          (MM)       STA  END
_____________________________________________________________________

  1         58.     2 - 16MM          0.          6485.         YES  YES
  2        442.     2 - 16MM          0.          6485.         YES  YES


        B E A M  N O.   575 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    3.58 KNS  Vc=  103.48 KNS  Vs=    0.00 KNS
 Tu=    7.83 KN-MET  Tc=    4.0 KN-MET  Ts=   10.4 KN-MET  LOAD    27
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2803. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.58 SQ.CM.

 AT END   SUPPORT - Vu=    8.41 KNS  Vc=  104.70 KNS  Vs=    0.00 KNS
 Tu=    1.05 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                  STIRRUPS ARE NOT REQUIRED.
```



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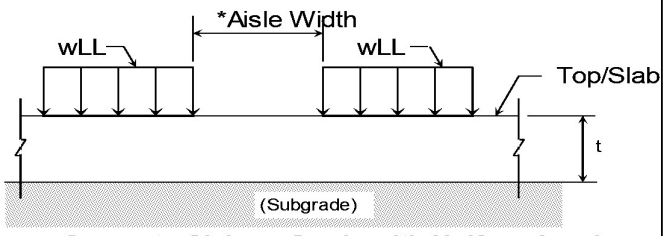
9. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

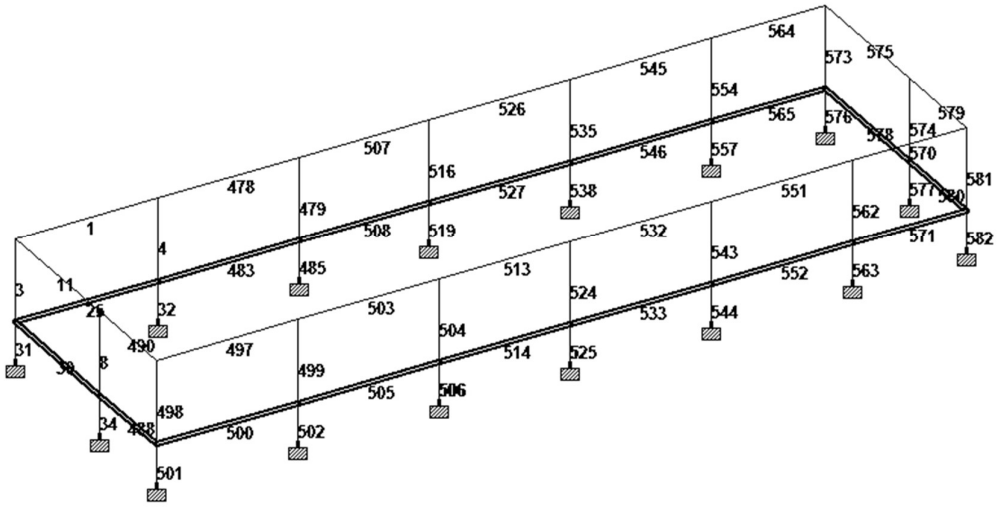
Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

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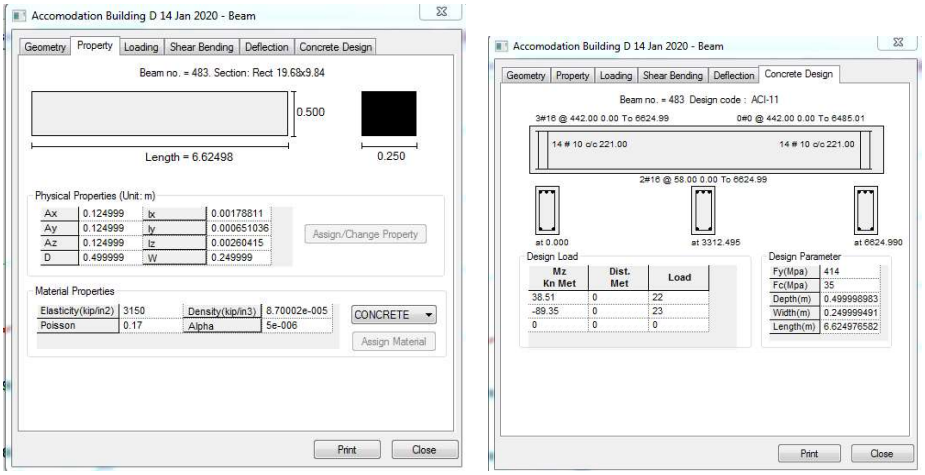
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10. DESIGN OF BEAM-(GB1)



MOMENT ENVELOPE FOR BEAMS GB1



Maximum Moment required from Analysis = 54 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-50-10-16/2=432mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **343 mm^2**

Ast.min= $$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **48 Knm (Sagging)**

Hence Provide 4-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

Ast.pro= **600mm^2 >>>>>>(Ast.min, Ast.req)**

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STAAD OUTPUT FOR GB1

ACI 318-11 BEAM NO. 483 DESIGN RESULTS

LEN - 6625. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	6625.	YES	YES
2	442.	3 - 16MM	0.	6625.	YES	YES

B E A M N O. 483 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 43.08 KNS Vc= 104.00 KNS Vs= 0.00 KNS
Tu= 0.38 KN-MET Tc= 3.7 KN-MET Ts= 0.0 KN-MET LOAD 3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2872. MM

AT END SUPPORT - Vu= 42.82 KNS Vc= 104.00 KNS Vs= 0.00 KNS
Tu= 0.38 KN-MET Tc= 3.7 KN-MET Ts= 0.0 KN-MET LOAD 3
NO STIRRUPS ARE REQUIRED FOR TORSION.

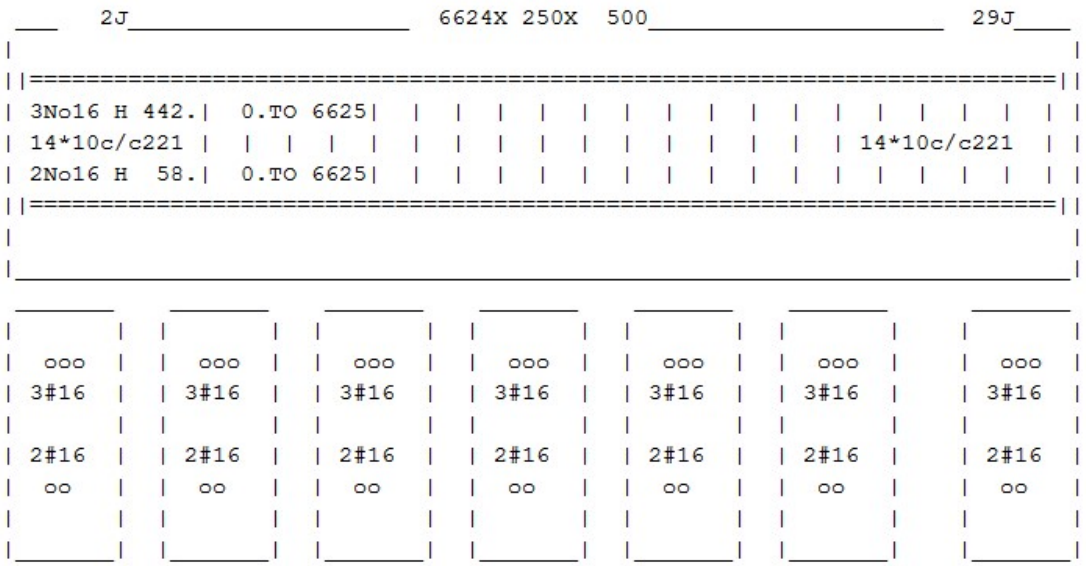
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```
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2872. MM

 AT END   SUPPORT - Vu=   42.82 KNS  Vc=  104.00 KNS  Vs=    0.00 KNS
 Tu=    0.38 KN-MET  Tc=    3.7 KN-MET  Ts=    0.0 KN-MET  LOAD     3
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 2872. MM
```

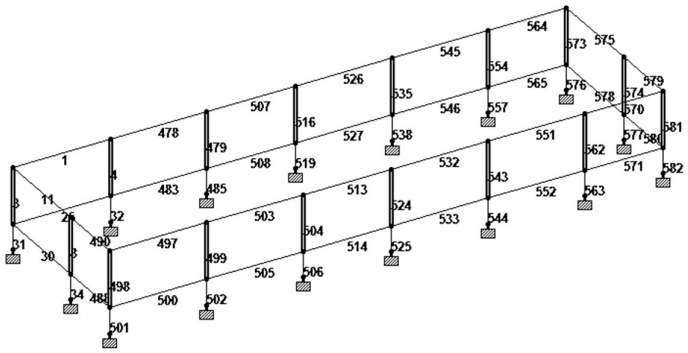


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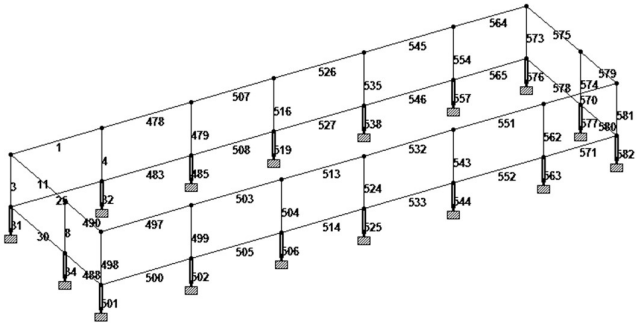
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11. CONCRETE COLUMN DESIGN



Column numbers C1



Column numbers CR

Beam End Force Summary

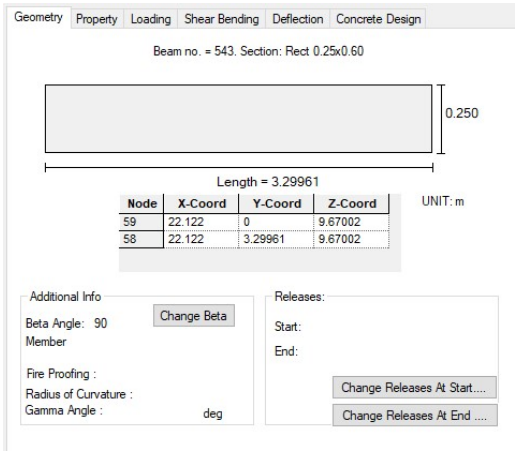
The signs of the forces at end B of each beam have been reversed. For example: this means that the Min Fx entry gives the largest value for an beam.

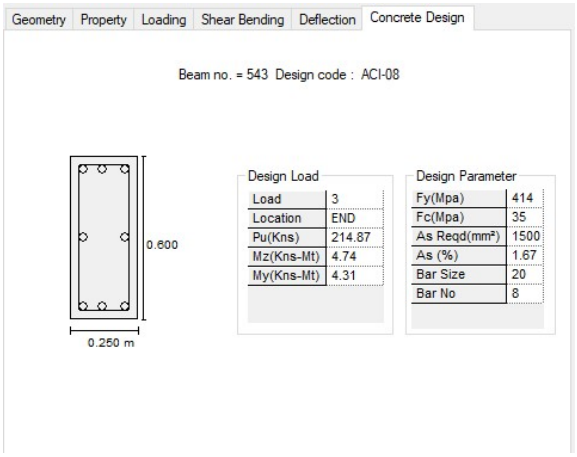
				Axial	Shear		Torsion	Bending	
	Beam	Node	L/C	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kNm)	My (kNm)	Mz (kNm)
Max Fx	502	39	22:GENERATE	**466.112**	-0.312	5.921	0.913	4.015	-4.097
Min Fx	527	44	23:GENERATE	**-47.917**	50.291	0.025	-0.146	-0.029	48.028
Max Fy	478	5	23:GENERATE	54.066	**156.936**	0.016	4.438	0.164	163.198
Min Fy	497	37	22:GENERATE	31.979	**-166.783**	-0.450	1.633	-1.554	168.171
Max Fz	34	7	27:GENERATE	103.904	19.085	**33.788**	4.032	-14.337	-4.906
Min Fz	581	77	23:GENERATE	146.744	-4.839	**-44.031**	-1.815	60.410	-9.997
Max Mx	498	34	27:GENERATE	161.736	-29.038	-13.341	**17.913**	29.859	-32.926
Min Mx	497	36	27:GENERATE	40.912	124.732	-5.431	**-19.633**	14.120	64.423
Max My	581	77	23:GENERATE	146.744	-4.839	-44.031	-1.815	**60.410**	-9.997
Min My	581	76	23:GENERATE	129.912	-4.839	-44.031	-1.815	**-84.875**	5.968
Max Mz	497	37	22:GENERATE	31.979	-166.783	-0.450	1.633	-1.554	**168.171**
Min Mz	519	47	31:GENERATE	274.240	45.653	-3.457	1.608	-6.171	**-131.051**

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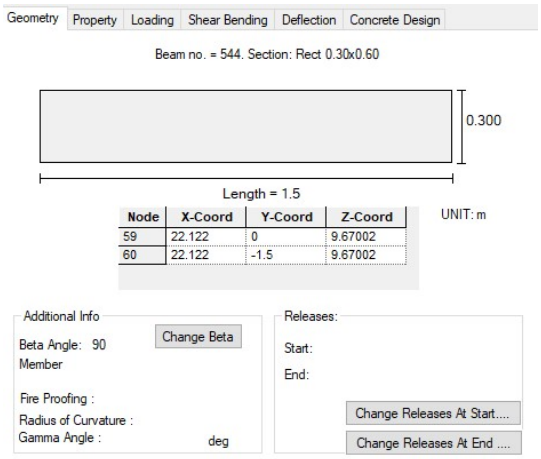
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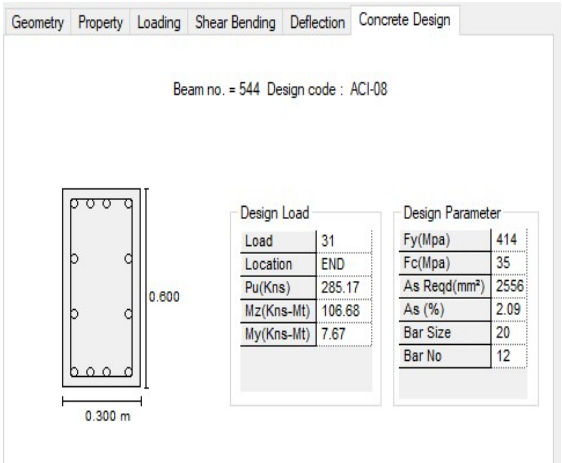
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Column Design-C1





Column Design-CR (Column Neck)

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11.1. Staad Output For Column C1 & Cr

```
      COLUMN   NO.    543   DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 600.0 X 250.0 MMS, TIED
         ONLY MINIMUM STEEL IS REQUIRED.
         AREA OF STEEL REQUIRED =   1500.0   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

  8 - 20 MM                 1.676         3       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 250.00 MM


      COLUMN   NO.    544   DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 600.0 X 300.0 MMS, TIED
         AREA OF STEEL REQUIRED =   2556.0   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

 12 - 20 MM                 2.094        31       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    10 SPACING 300.00 MM
```

Hence Provide 8-20 dia-Vertical Reinf for Column Reinforcement-C1 &
Provide 8-25 dia-Vertical Reinf for Column Reinforcement-CR(Column Neck)

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12. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r"	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db" = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}*f_y/(0.85*f'_c*b)$	6.61	mm					
Section capacity ΦMn = $\Phi*A_{s\ t}*f_y*(d-a/2)$	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'^{2}	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b **$l_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $l_e/240$ 3.333 mm > 0.079 mm O.K

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13. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 10: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES:

BUILDING PLANS	MD-216-6000-EG-AR-GAD-3087
OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-6000-EG-CV-DED-1481
LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-6000-EG-CV-DED-1482
GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

STAAD SPACE

START JOB INFORMATION

ENGINEER DATE 18-Oct-19

END JOB INFORMATION

INPUT WIDTH 79

UNIT METER KN

JOINT COORDINATES

1 0.122 0 0; 2 5.77201 0 0; 4 0.122 3.29961 0; 5 5.77201 3.29961 0;

7 0.122 0 5.78501; 10 0.122 3.29961 5.78501; 23 0.122 -1.5 0;

24 5.77201 -1.5 0; 26 0.122 -1.5 5.78501; 28 11.372 3.29961 0; 29 11.372 0 0;

32 11.372 -1.5 0; 34 0.122 0 9.67002; 35 5.77201 0 9.67002;

36 0.122 3.29961 9.67002; 37 5.77201 3.29961 9.67002; 38 0.122 -1.5 9.67002;

39 5.77201 -1.5 9.67002; 40 11.372 3.29961 9.67002; 41 11.372 0 9.67002;

42 11.372 -1.5 9.67002; 43 16.522 3.29961 0; 44 16.522 0 0; 47 16.522 -1.5 0;

49 16.522 3.29961 9.67002; 50 16.522 0 9.67002; 51 16.522 -1.5 9.67002;

52 22.122 3.29961 0; 53 22.122 0 0; 56 22.122 -1.5 0;

58 22.122 3.29961 9.67002; 59 22.122 0 9.67002; 60 22.122 -1.5 9.67002;

61 27.7221 3.29961 0; 62 27.7221 0 0; 65 27.7221 -1.5 0;

67 27.7221 3.29961 9.67002; 68 27.7221 0 9.67002; 69 27.7221 -1.5 9.67002;

70 32.2721 3.29961 0; 71 32.2721 0 0; 72 32.2721 3.29961 5.78501;

73 32.2721 0 5.78501; 74 32.2721 -1.5 0; 75 32.2721 -1.5 5.78501;

76 32.2721 3.29961 9.67002; 77 32.2721 0 9.67002; 78 32.2721 -1.5 9.67002;

MEMBER INCIDENCES

1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;

34 7 26; 478 5 28; 479 29 28; 483 2 29; 485 29 32; 488 7 34; 490 10 36;

497 36 37; 498 34 36; 499 35 37; 500 35 34; 501 34 38; 502 35 39; 503 37 40;

504 41 40; 505 41 35; 506 41 42; 507 28 43; 508 29 44; 513 40 49; 514 41 50;

516 44 43; 519 44 47; 524 50 49; 525 50 51; 526 43 52; 527 44 53; 532 49 58;

533 50 59; 535 53 52; 538 53 56; 543 59 58; 544 59 60; 545 52 61; 546 53 62;

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551 58 67; 552 59 68; 554 62 61; 557 62 65; 562 68 67; 563 68 69; 564 61 70;

565 62 71; 570 67 76; 571 68 77; 573 71 70; 574 73 72; 575 70 72; 576 71 74;

577 73 75; 578 71 73; 579 72 76; 580 73 77; 581 77 76; 582 77 78;

DEFINE MATERIAL START

ISOTROPIC CONCRETE

E 2.17184e+007

POISSON 0.17

DENSITY 23.6158

ALPHA 5e-006

DAMP 0.05

TYPE CONCRETE

STRENGTH FCU 27578.9

END DEFINE MATERIAL

MEMBER PROPERTY AMERICAN

31 32 34 485 501 502 506 519 525 538 544 557 563 576 577 582 PRIS YD 0.3 ZD 0.6

1 478 497 503 507 513 526 532 545 551 564 570 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 -

580 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

11 490 575 579 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

3 4 8 479 498 499 504 516 524 535 543 554 562 573 574 581 PRIS YD 0.25 ZD 0.6

CONSTANTS

BETA 90 MEMB 4 32 479 485 499 502 504 506 516 519 524 525 535 538 543 544 -

554 557 562 563

MATERIAL CONCRETE ALL

SUPPORTS

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23 24 26 32 38 39 42 47 51 56 60 65 69 74 75 78 FIXED

MEMBER CRACKED

3 4 8 31 32 34 REDUCTION RIY 0.700000 RIZ 0.700000

1 11 25 30 REDUCTION RIY 0.350000 RIZ 0.350000

MEMBER OFFSET

478 503 507 513 526 532 545 551 564 570 START 0.3 0 0

1 478 497 503 507 513 526 532 545 551 END -0.3 0 0

11 490 575 579 START 0 0 0.3

11 490 575 579 END 0 0 -0.3

1 497 START 0.125 0 0

564 570 END -0.125 0 0

DEFINE IBC 2003

SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 CT 0.04

SELFWEIGHT 1

MEMBER WEIGHT

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 UNI 12

1 478 497 503 507 513 526 532 545 551 564 570 UNI 35

DEFINE WIND LOAD

TYPE 1 WIND X WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 3.500 M 33.000 M 10.000 M 2.000 0.010 0 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 0 1 0.850 0.800 -0.550

!> END GENERATED DATA BLOCK

INT 1.59699 1.59699 1.59699 HEIG 0 4.572 4.572

EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -

75 78

TYPE 2 WIND X LEEWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

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1 3.500 M 33.000 M 10.000 M 2.000 0.010 1 -

0 0 0 0 1.030 1.000 1.000 0.850 0 -

1 0 1 0.850 -0.200 0.550

!> END GENERATED DATA BLOCK

INT -0.934826 -0.934826 -0.945969 HEIG 0 3.5 3.5

EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -

75 78

TYPE 3 WIND Z WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 6.000 M 10.000 M 2.000 0.010 0 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 1 1 0.850 0.800 -0.550

!> END GENERATED DATA BLOCK

INT 1.59699 1.59699 1.64086 HEIG 0 4.572 4.572

EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -

75 78

TYPE 4 WIND Z LEEWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 6.000 M 10.000 M 2.000 0.010 1 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 0 1 0.850 -0.500 0.550

!> END GENERATED DATA BLOCK

INT -1.26591 -1.26591 -1.29332 HEIG 0 4 4

EXP 0.8 JOINT 1 2 4 5 7 10 23 24 26 28 29 32 38 39 42 47 51 56 60 65 69 74 -

75 78

LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z

IBC LOAD Z 1

PERFORM ANALYSIS

CHANGE

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LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X

IBC LOAD X 1

PERFORM ANALYSIS

CHANGE

LOAD 3 LOADTYPE Dead TITLE DL

SELFWEIGHT Y -1

MEMBER LOAD

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 UNI GY -12

1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -35

11 490 UMOM GZ 1

575 579 UMOM GZ -1

1 478 507 526 545 564 UMOM GX -1

497 503 513 532 551 570 UMOM GX 1

LOAD 4 LOADTYPE Roof Live TITLE LL

MEMBER LOAD

1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -2.42

11 490 UMOM GZ 0.1

575 579 UMOM GZ -0.1

1 478 507 526 545 564 UMOM GX -0.1

497 503 513 532 551 570 UMOM GX 0.1

LOAD 5 LOADTYPE Wind TITLE WL+X

WIND LOAD X 1 TYPE 1 XR 0 0.2 YR 0 4 ZR 0 10

WIND LOAD -X 1 TYPE 2 XR 32.272 33 YR 0 4

WIND LOAD Z -1 TYPE 3 XR 0 33 YR 0 4 ZR 0 0.1

WIND LOAD -Z 1 TYPE 4 YR 0 4 ZR 9.67 10

LOAD 6 LOADTYPE Wind TITLE WL+Z

WIND LOAD Z 1 TYPE 3 XR 0 33 YR 0 4 ZR 0 0.1

WIND LOAD -Z 1 TYPE 4 YR 0 4 ZR 9.67 10

WIND LOAD X 1 TYPE 1 XR 0.1 0.122 YR 0 4 ZR -0.6 10

WIND LOAD -X 1 TYPE 2 XR 32.272 33 YR 0 4 ZR -0.6 10

LOAD 7 LOADTYPE Temperature TITLE TEMP

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TEMPERATURE LOAD

1 3 4 8 11 478 479 490 497 TO 499 503 504 507 513 516 524 526 532 535 543
-

545 551 554 562 564 570 573 TO 575 579 581 TEMP 25

LOAD COMB 8 SLS: (DL)

3 1.0

LOAD COMB 9 SLS: (DL+T)

3 1.0 7 1.0

LOAD COMB 10 SLS: (D+LR)

3 1.0 4 1.0

LOAD COMB 11 SLS: (D+0.75LR)

3 1.0 4 0.75

LOAD COMB 12 SLS: (D+0.7EZ)

3 1.0 1 0.7

LOAD COMB 13 SLS: (D+0.7EX)

3 1.0 2 0.7

LOAD COMB 14 SLS: (D+0.75WX+0.75LR)

3 1.0 5 0.75 4 0.75

LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)

3 1.0 6 0.75 4 0.75

LOAD COMB 16 SLS: (0.6D+WX)

3 0.6 5 1.0

LOAD COMB 17 SLS: (0.6D+WZ)

3 0.6 6 1.0

LOAD COMB 18 SLS: (0.6D+0.7EX)

3 0.6 2 0.7

LOAD COMB 19 SLS: (0.6D+0.7EZ)

3 0.6 1 0.7

LOAD COMB 20 SLS: (D+WX)

3 1.0 5 1.0

LOAD COMB 21 SLS: (D+WZ)

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3 1.0 6 1.0

LOAD COMB 22 ULS: (1.4 DL)

3 1.4

LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)

3 1.2 5 0.8 4 1.6

LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 6 1.6 4 0.5

LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)

3 1.2 4 0.5 5 1.6

LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 4 0.5 6 1.6

LOAD COMB 29 ULS: (1.2D+EX)

3 1.2 2 1.0

LOAD COMB 30 ULS: (1.2D+EZ)

3 1.2 1 1.0

LOAD COMB 31 ULS: (0.90D+1.6WX)

3 0.9 5 1.6

LOAD COMB 32 ULS: (0.90D+1.6WZ)

3 0.9 6 1.6

LOAD COMB 33 ULS: (0.90D+EZ)

3 0.9 1 1.0

LOAD COMB 34 ULS: (0.90D+EX)

3 0.9 2 1.0

PERFORM ANALYSIS

START CONCRETE DESIGN

CODE ACI

FC 35000 ALL

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CLB 0.04 ALL

CLS 0.04 ALL

CLT 0.04 ALL

MAXMAIN 20 MEMB 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 -

525 535 538 543 544 554 557 562 563 573 574 576 577 581 582

MINMAIN 20 MEMB 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 -

525 535 538 543 544 554 557 562 563 573 574 576 577 581 582

MAXMAIN 25 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579

MINMAIN 20 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579

MAXMAIN 16 MEMB 11 490 575 579

MINMAIN 16 MEMB 11 490 575 579

MAXMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580

MINMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580

MINSEC 10 ALL

EFACE 0.74 MEMB 1 478 497 503 507 513 526 532 545 551

SFACE 0.74 MEMB 478 503 507 513 526 532 545 551 564 570

EFACE 0.74 MEMB 11 490 575 579

SFACE 0.74 MEMB 11 490 575 579

EFACE 0.69 MEMB 564 570

SFACE 0.69 MEMB 1 497

DESIGN BEAM 1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 527 532 533 545 546 551 552 564 565 570 571 575 578 TO 580

DESIGN COLUMN 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 525 535 538 543 544 554 557 562 563 573 574 576 577 581 582

END CONCRETE DESIGN

FINISH

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ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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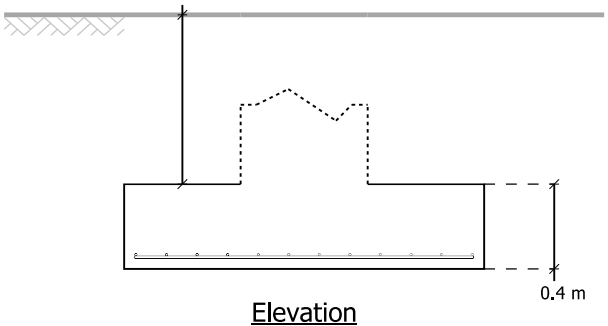
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Isolated Footing Design(ACI 318-08)

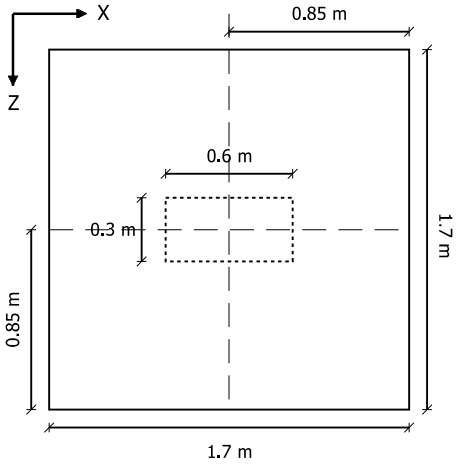
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
32	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
32	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1-GRID-1 & C



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	SL:1.000 x DL	1.00	1.00	1.00
102	SL:1.000 x DL+1.000 x LL	1.00	1.00	1.00
105	SL:1.000 x DL+0.750 x LL	1.00	1.00	1.00
107	SL:1.000 x DL+1.000 x WLX	1.00	1.00	1.00
109	SL:1.000 x DL+1.000 x WLZ	1.00	1.00	1.00
111	SL:1.000 x DL+0.700 x ELX	1.00	1.00	1.00
113	SL:1.000 x DL+0.700 x ELZ	1.00	1.00	1.00
115	SL:1.000 x DL+0.750 x LL+0.750 x WLX	1.00	1.00	1.00
119	SL:1.000 x DL+0.750 x LL+0.750 x WLZ	1.00	1.00	1.00
123	SL:1.000 x DL+0.750 x LL+0.525 x ELX	1.00	1.00	1.00
127	SL:1.000 x DL+0.750 x LL+0.525 x ELZ	1.00	1.00	1.00
131	SL:0.600 x DL+1.000 x WLX	1.00	1.00	0.60
133	SL:0.600 x DL+1.000 x WLZ	1.00	1.00	0.60
135	SL:0.600 x DL+0.700 x ELX	1.00	1.00	0.60
137	SL:0.600 x DL+0.700 x ELZ	1.00	1.00	0.60

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	UL:1.400 x DL	1.00	1.00	1.40
202	UL:1.200 x DL+1.600 x LL	1.00	1.00	1.20
204	UL:1.200 x DL+1.000 x LL	1.00	1.00	1.20
206	UL:1.200 x DL+0.800 x WLX	1.00	1.00	1.20
208	UL:1.200 x DL+0.800 x WLZ	1.00	1.00	1.20
214	UL:1.200 x DL+1.000 x LL+1.600 x WLX	1.00	1.00	1.20
218	UL:1.200 x DL+1.000 x LL+1.600 x WLZ	1.00	1.00	1.20
222	UL:1.200 x DL+1.000 x LL+1.000 x ELX	1.00	1.00	1.20
224	UL:1.200 x DL+1.000 x LL+1.000 x ELZ	1.00	1.00	1.20
226	UL:0.900 x DL+1.600 x WLX	1.00	1.00	0.90
228	UL:0.900 x DL+1.600 x WLZ	1.00	1.00	0.90
230	UL:0.900 x DL+1.000 x ELX	1.00	1.00	0.90
232	UL:0.900 x DL+1.000 x ELZ	1.00	1.00	0.90

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	303.187	-1.028	0.018	-4.740	1.262
102	316.099	-1.441	0.018	-5.225	1.559
105	312.871	-1.337	0.018	-5.104	1.485
107	302.977	3.030	-24.596	-80.412	-4.554
109	303.033	2.636	24.651	71.024	-3.907
111	302.628	7.154	0.018	-4.746	-11.941
113	303.195	-1.066	7.155	24.672	1.326
115	312.713	1.706	-18.442	-61.858	-2.877
119	312.756	1.411	18.493	51.719	-2.392
123	312.452	4.799	0.018	-5.108	-8.417
127	312.877	-1.366	5.371	16.955	1.533
131	181.702	3.441	-24.603	-78.516	-5.059
133	181.759	3.047	24.644	72.920	-4.412
135	181.353	7.565	0.011	-2.850	-12.446
137	181.920	-0.655	7.147	26.568	0.822

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	424.462	-1.439	0.026	-6.636	1.766

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
202	384.484	-1.894	0.022	-6.464	1.990
204	376.737	-1.646	0.022	-6.173	1.811
206	363.656	2.012	-19.669	-66.225	-3.138
208	363.702	1.698	19.728	54.923	-2.621
214	376.400	4.846	-39.361	-127.248	-7.494
218	376.491	4.216	39.434	115.049	-6.459
222	375.938	10.042	0.022	-6.181	-17.050
224	376.748	-1.701	10.217	35.844	1.904
226	272.532	5.567	-39.366	-125.341	-8.169
228	272.623	4.937	39.429	116.957	-7.135
230	272.069	10.763	0.017	-4.274	-17.726
232	272.879	-0.980	10.211	37.751	1.228

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.907 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 101

Width (W_2) = 1.700 m Governing Load Case : # 101

Depth (D_2) = 0.400 m Governing Load Case : # 214

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

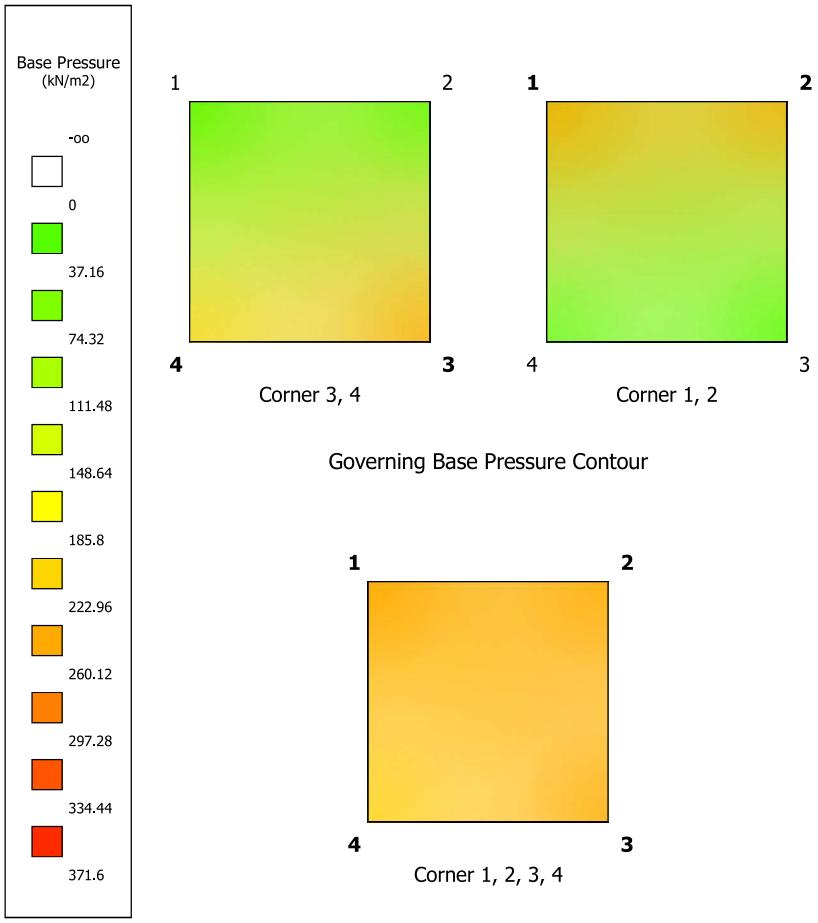
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 39.023 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



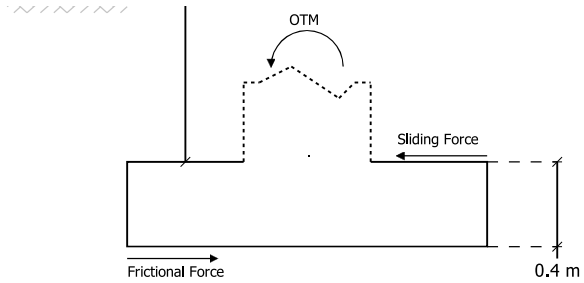
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
107	**231.1166**	245.1996	24.7604	10.6774	0.000
107	231.1166	**245.1996**	24.7604	10.6774	0.000
109	23.1167	35.2359	**232.7997**	220.6805	0.000
109	23.1167	35.2359	232.7997	**220.6805**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
107	**231.1166**	245.1996	24.7604	10.6774
107	231.1166	**245.1996**	24.7604	10.6774
109	23.1167	35.2359	**232.7997**	220.6805
109	23.1167	35.2359	232.7997	**220.6805**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	208.728	11688.314	295.139	1.500	66.793	187.969	1.500
102	154.143	12085.827	217.973	1.500	62.685	152.413	1.500
105	164.632	11986.513	232.803	1.500	63.638	159.776	1.500
107	70.787	8.719	12.238	1.500	3.501	54.508	1.500
109	81.368	8.701	12.235	1.500	3.907	63.350	1.500
111	29.948	11648.283	42.352	1.500	66.613	21.211	1.500
113	201.212	29.991	41.951	1.500	11.481	179.389	1.500
115	129.039	11.934	16.806	1.500	4.683	90.621	1.500
119	156.058	11.903	16.785	1.500	5.485	109.116	1.500
123	45.831	11956.067	64.814	1.500	63.516	31.182	1.500
127	161.162	41.000	56.193	1.500	16.978	155.166	1.500
131	39.253	5.489	7.688	1.500	2.251	30.758	1.500
133	44.333	5.482	7.694	1.500	2.404	35.159	1.500
135	17.826	12204.162	25.209	1.500	69.808	12.773	1.500
137	206.318	18.914	26.637	1.500	6.766	182.795	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 135
Governing Disturbing Force : 7.565kN
Governing Restoring Force : 134.856kN
Minimum Sliding Ratio for the Critical Load Case : 17.826
Critical Load Case for Overturning about X-Direction : 131
Governing Overturning Moment : -88.357kNm
Governing Resisting Moment : 198.917kNm
Minimum Overturning Ratio for the Critical Load Case : 2.251

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 133
Governing Disturbing Force : 24.644kN
Governing Restoring Force : 135.091kN
Minimum Sliding Ratio for the Critical Load Case : 5.482
Critical Load Case for Overturning about Z-Direction : 135
Governing Overturning Moment : -15.472kNm
Governing Resisting Moment : 197.629kNm
Minimum Overturning Ratio for the Critical Load Case : 12.773

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 131

Governing Disturbing Force : 24.843kN

Governing Restoring Force : 191.001kN

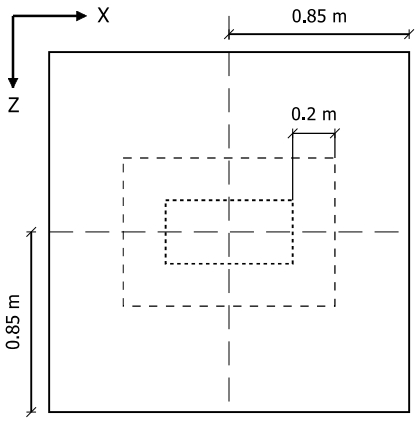
Minimum Sliding Ratio for the Critical Load Case : 7.688

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 408.982kN, Load Case # 201

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.656kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2146.802kN

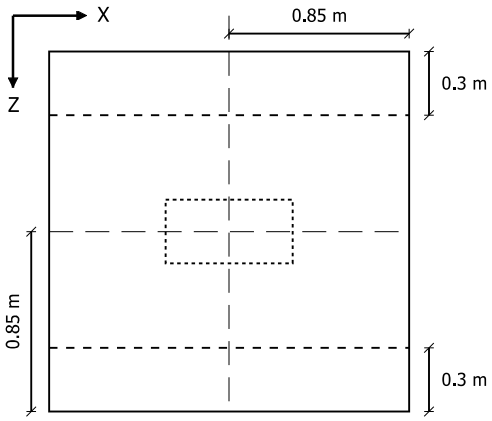
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.102kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

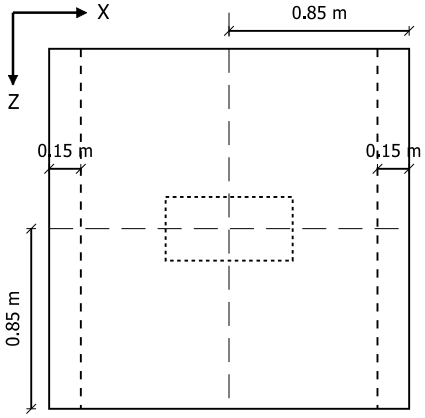
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 214 $V_{ux} = V_{ux}|_{x=D_x}$ = 179.156 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.206 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

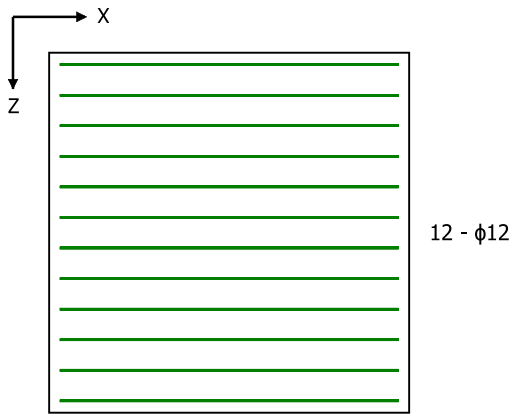
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 201 $V_{uz} = V_{uz}|_{z=D_z}$ = 63.642 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 201

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00166
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.550 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	46.657 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	51.841 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00062
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00053
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1131.328 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.500 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

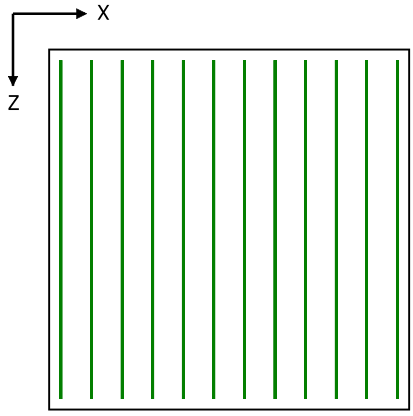
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.200

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00172

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.700 m

Ultimate moment, $M_u|_{x=D_x}$ = 118.542 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 131.714 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00169

(Based on gross depth) ρ x d_{eff} / Depth = 0.00141

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1170.668 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.650 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

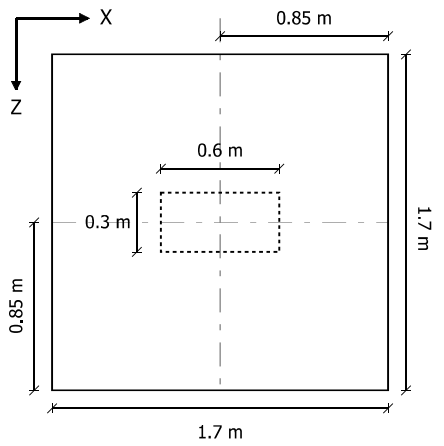
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

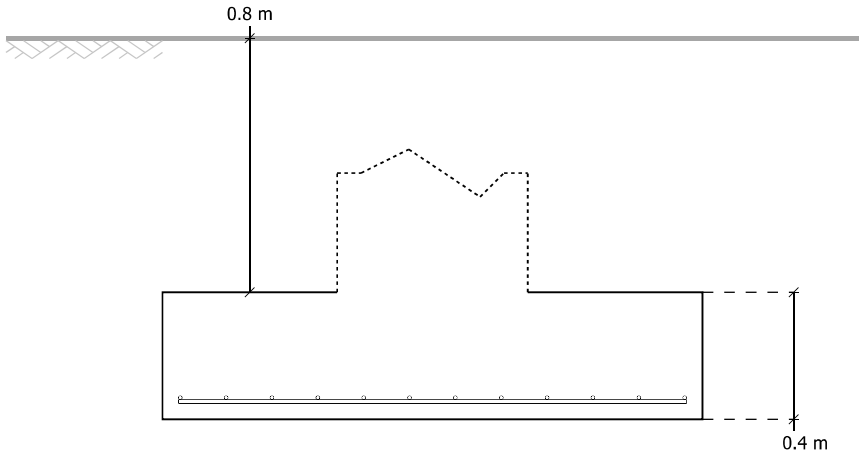
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.159

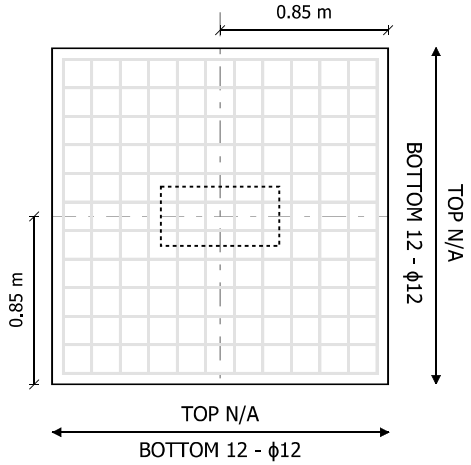
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

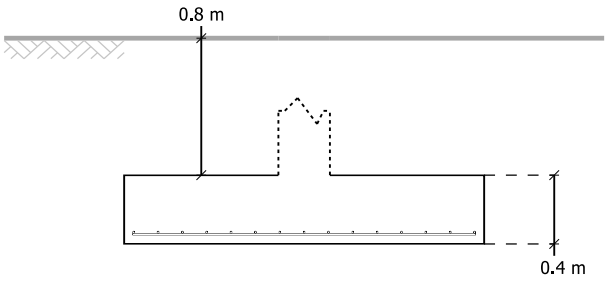
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

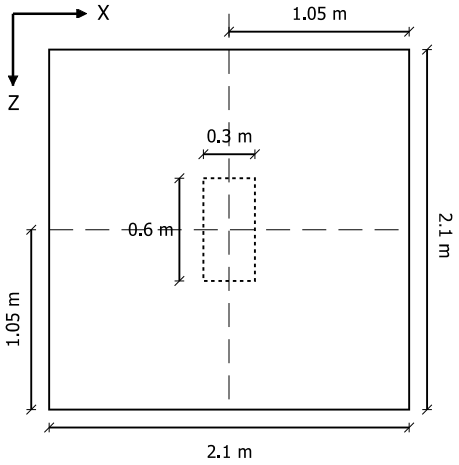
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
26	1	2.100m	2.100m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
26	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2-GRID-2 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2100.000mm
Footing Width - Z (Fw) : 2100.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.100m

Initial Width (W_o) = 2.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	SL:1.000 x DL	1.00	1.00	1.00
102	SL:1.000 x DL+1.000 x LL	1.00	1.00	1.00
105	SL:1.000 x DL+0.750 x LL	1.00	1.00	1.00
107	SL:1.000 x DL+1.000 x WLX	1.00	1.00	1.00
109	SL:1.000 x DL+1.000 x WLZ	1.00	1.00	1.00
111	SL:1.000 x DL+0.700 x ELX	1.00	1.00	1.00
113	SL:1.000 x DL+0.700 x ELZ	1.00	1.00	1.00
115	SL:1.000 x DL+0.750 x LL+0.750 x WLX	1.00	1.00	1.00
119	SL:1.000 x DL+0.750 x LL+0.750 x WLZ	1.00	1.00	1.00
123	SL:1.000 x DL+0.750 x LL+0.525 x ELX	1.00	1.00	1.00
127	SL:1.000 x DL+0.750 x LL+0.525 x ELZ	1.00	1.00	1.00
131	SL:0.600 x DL+1.000 x WLX	1.00	1.00	0.60
133	SL:0.600 x DL+1.000 x WLZ	1.00	1.00	0.60
135	SL:0.600 x DL+0.700 x ELX	1.00	1.00	0.60
137	SL:0.600 x DL+0.700 x ELZ	1.00	1.00	0.60

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	UL:1.400 x DL	1.00	1.00	1.40
202	UL:1.200 x DL+1.600 x LL	1.00	1.00	1.20
204	UL:1.200 x DL+1.000 x LL	1.00	1.00	1.20
206	UL:1.200 x DL+0.800 x WLX	1.00	1.00	1.20
208	UL:1.200 x DL+0.800 x WLZ	1.00	1.00	1.20
214	UL:1.200 x DL+1.000 x LL+1.600 x WLX	1.00	1.00	1.20
218	UL:1.200 x DL+1.000 x LL+1.600 x WLZ	1.00	1.00	1.20
222	UL:1.200 x DL+1.000 x LL+1.000 x ELX	1.00	1.00	1.20
224	UL:1.200 x DL+1.000 x LL+1.000 x ELZ	1.00	1.00	1.20
226	UL:0.900 x DL+1.600 x WLX	1.00	1.00	0.90
228	UL:0.900 x DL+1.600 x WLZ	1.00	1.00	0.90
230	UL:0.900 x DL+1.000 x ELX	1.00	1.00	0.90
232	UL:0.900 x DL+1.000 x ELZ	1.00	1.00	0.90

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	109.651	-0.736	11.721	5.915	-0.415
102	109.667	-0.744	11.726	5.934	-0.412
105	109.663	-0.742	11.725	5.929	-0.412
107	108.543	13.134	11.578	5.502	-23.869
109	97.114	11.754	24.041	24.171	-21.066
111	109.635	0.555	11.713	5.894	-3.129
113	103.083	-0.250	17.243	14.716	-1.489
115	108.832	9.661	11.618	5.619	-18.003
119	100.261	8.625	20.965	19.621	-15.900
123	109.651	0.226	11.719	5.913	-2.448
127	104.738	-0.378	15.867	12.529	-1.218
131	64.682	13.429	6.890	3.136	-23.703
133	53.254	12.048	19.352	21.805	-20.900
135	65.775	0.849	7.025	3.528	-2.963
137	59.223	0.044	12.555	12.349	-1.323

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	153.511	-1.031	16.409	8.282	-0.581

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
202	131.607	-0.896	14.074	7.128	-0.493
204	131.597	-0.891	14.071	7.117	-0.495
206	130.694	10.213	13.951	6.768	-19.261
208	121.552	9.109	23.921	21.703	-17.018
214	129.825	21.302	13.843	6.455	-38.021
218	111.539	19.093	33.782	36.326	-33.536
222	131.575	0.953	14.059	7.087	-4.372
224	122.215	-0.197	21.960	19.688	-2.029
226	96.913	21.530	10.321	4.662	-37.900
228	78.627	19.321	30.260	34.533	-33.415
230	98.663	1.182	10.537	5.294	-4.251
232	89.303	0.031	18.438	17.895	-1.908

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.410 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.997 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.100 m Governing Load Case : # 101

Width (W_2) = 2.100 m Governing Load Case : # 101

Depth (D_2) = 0.400 m Governing Load Case : # 218

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.410 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 42.335 kN

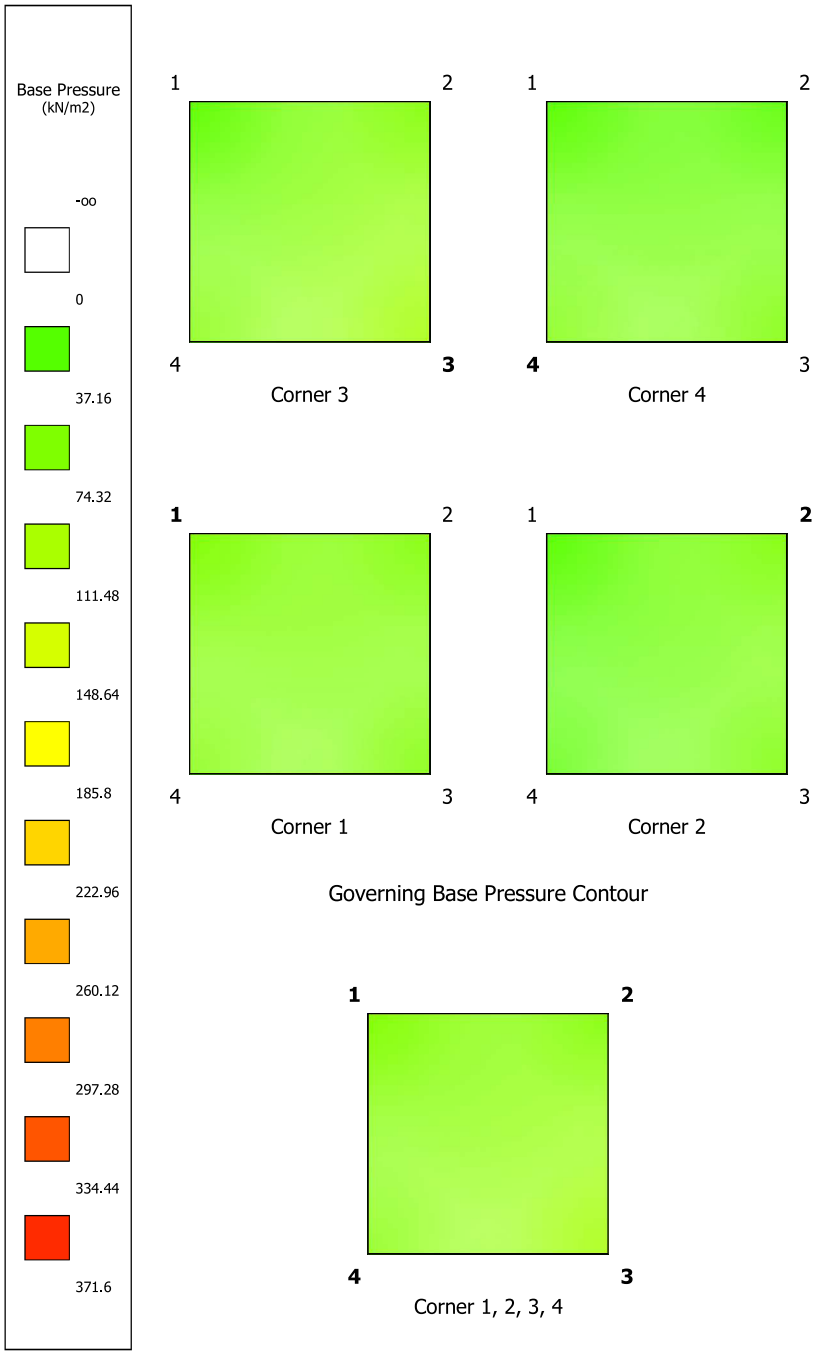
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 60.910 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



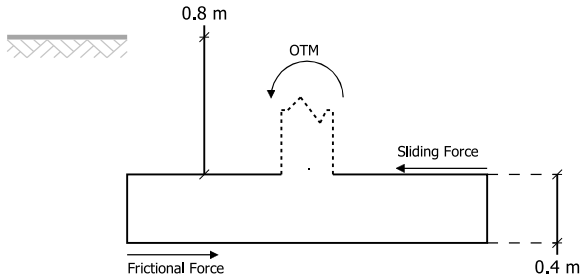
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**41.3275**	41.4835	55.2235	55.0676	0.000
107	22.5908	**60.3273**	73.4577	35.7211	0.000
109	6.8480	40.2365	**84.0176**	50.6291	0.000
113	31.8837	33.6836	61.6888	**59.8889**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**41.3275**	41.4835	55.2235	55.0676
107	22.5908	**60.3273**	73.4577	35.7211
109	6.8480	40.2365	**84.0176**	50.6291
113	31.8837	33.6836	61.6888	**59.8889**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	167.729	10.535	14.869	1.500	21.081	1874.666	1.500
102	165.980	10.531	14.863	1.500	21.042	1979.600	1.500
105	166.414	10.532	14.865	1.500	21.052	1952.283	1.500
107	9.352	10.609	9.921	1.500	21.945	7.706	1.500
109	9.887	4.834	6.141	1.500	6.226	8.244	1.500
111	222.461	10.541	14.891	1.500	21.128	67.311	1.500
113	477.967	6.940	9.814	1.500	10.024	157.432	1.500
115	12.732	10.587	11.512	1.500	21.690	10.276	1.500
119	13.684	5.630	7.363	1.500	7.629	11.148	1.500
123	545.510	10.537	14.898	1.500	21.087	88.860	1.500
127	319.391	7.603	10.749	1.500	11.569	206.559	1.500
131	6.201	12.085	7.802	1.500	25.585	5.227	1.500
133	6.361	3.960	4.754	1.500	4.696	5.442	1.500
135	98.765	11.944	16.769	1.500	23.964	46.356	1.500
137	1816.346	6.380	9.023	1.500	8.348	109.052	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 131
Governing Disturbing Force : 13.429kN
Governing Restoring Force : 83.266kN
Minimum Sliding Ratio for the Critical Load Case : 6.201
Critical Load Case for Overturning about X-Direction : 133
Governing Overturning Moment : 29.546kNm
Governing Resisting Moment : 138.739kNm
Minimum Overturning Ratio for the Critical Load Case : 4.696

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 133
Governing Disturbing Force : 19.352kN
Governing Restoring Force : 76.638kN
Minimum Sliding Ratio for the Critical Load Case : 3.960
Critical Load Case for Overturning about Z-Direction : 131
Governing Overturning Moment : -29.075kNm
Governing Resisting Moment : 151.963kNm
Minimum Overturning Ratio for the Critical Load Case : 5.227

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

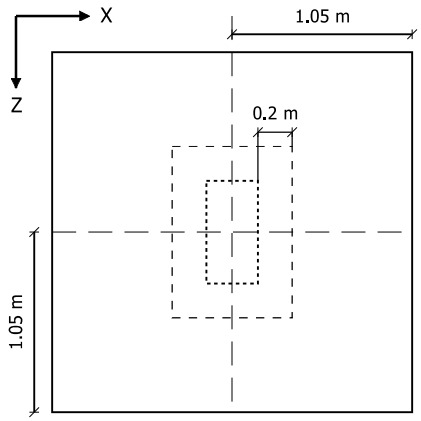
Critical Load Case for Sliding along Resultant Direction : 133
Governing Disturbing Force : 22.796kN
Governing Restoring Force : 108.383kN
Minimum Sliding Ratio for the Critical Load Case : 4.754

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 256.965kN, Load Case # 201

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.656kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

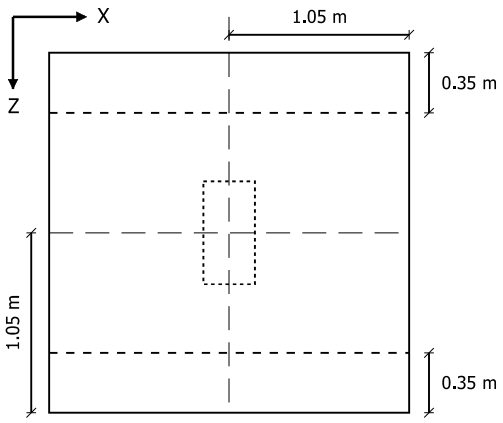
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.102kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.406m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

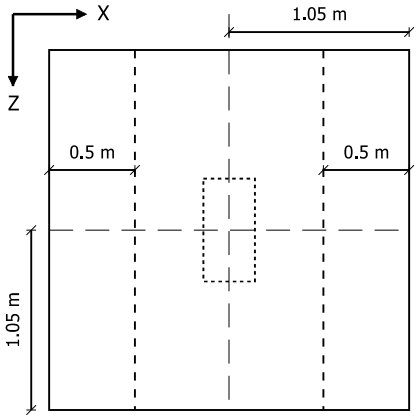
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{ux} is # 218 $V_{ux} = V_{ux}|_{x=D_x}$ = 67.690 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.556 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

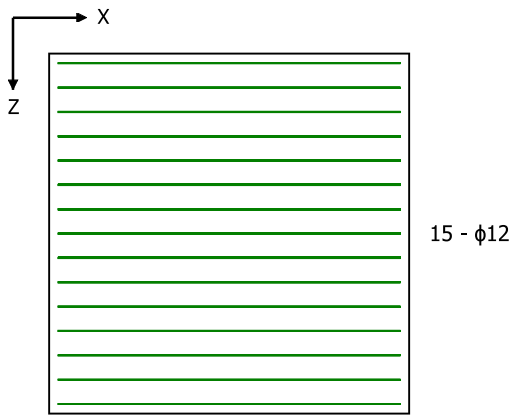
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{uz} is # 214 $V_{uz} = V_{uz}|_{z=D_z}$ = 93.061 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00164
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	67.263 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	74.736 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00072
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00062
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1380.565 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.850	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00235	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

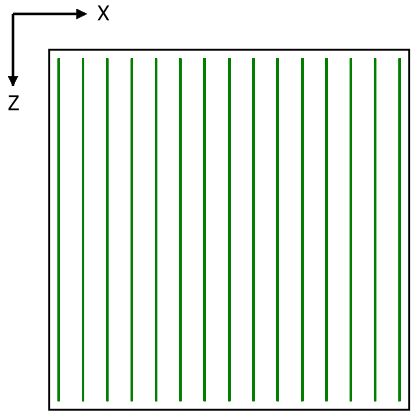
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.229

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 218

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00159	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	1.350	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	46.127	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	51.252	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00053	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00044	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1335.422	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times \left(L - D_{col}\right) - C_{cover}$ =	0.700	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

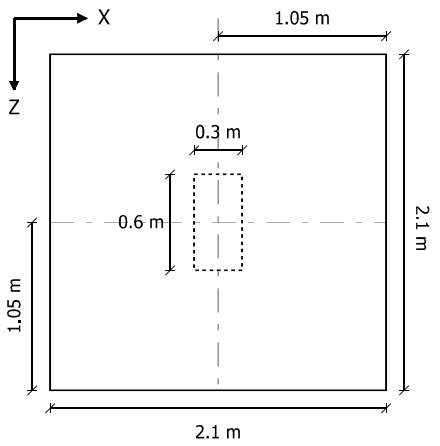
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.332	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ =	0.00243	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

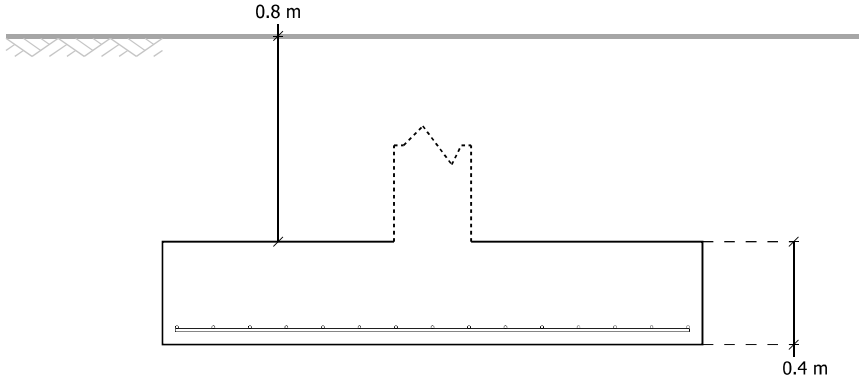
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.270

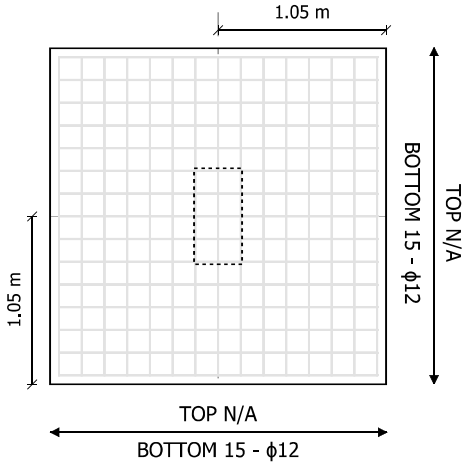
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

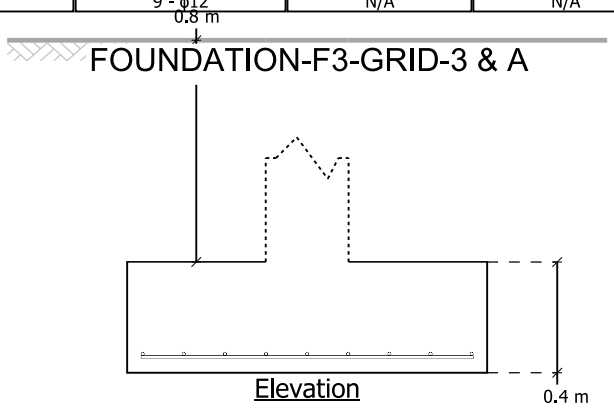
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.1 m	2.1 m	0.4 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

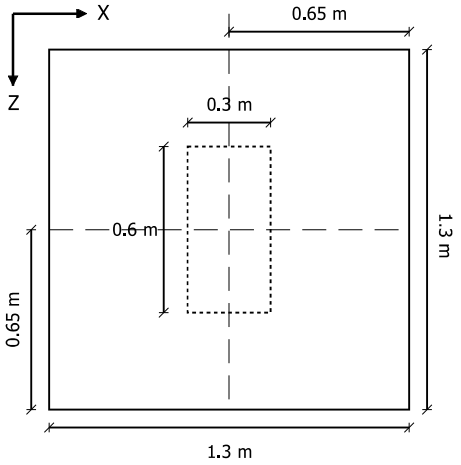
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
38	1	1.300m	1.300m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
38	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.300m

Initial Width (W_o) = 1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	SL:1.000 x DL	1.00	1.00	1.00
102	SL:1.000 x DL+1.000 x LL	1.00	1.00	1.00
105	SL:1.000 x DL+0.750 x LL	1.00	1.00	1.00
107	SL:1.000 x DL+1.000 x WLX	1.00	1.00	1.00
109	SL:1.000 x DL+1.000 x WLZ	1.00	1.00	1.00
111	SL:1.000 x DL+0.700 x ELX	1.00	1.00	1.00
113	SL:1.000 x DL+0.700 x ELZ	1.00	1.00	1.00
115	SL:1.000 x DL+0.750 x LL+0.750 x WLX	1.00	1.00	1.00
119	SL:1.000 x DL+0.750 x LL+0.750 x WLZ	1.00	1.00	1.00
123	SL:1.000 x DL+0.750 x LL+0.525 x ELX	1.00	1.00	1.00
127	SL:1.000 x DL+0.750 x LL+0.525 x ELZ	1.00	1.00	1.00
131	SL:0.600 x DL+1.000 x WLX	1.00	1.00	0.60
133	SL:0.600 x DL+1.000 x WLZ	1.00	1.00	0.60
135	SL:0.600 x DL+0.700 x ELX	1.00	1.00	0.60
137	SL:0.600 x DL+0.700 x ELZ	1.00	1.00	0.60

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	UL:1.400 x DL	1.00	1.00	1.40
202	UL:1.200 x DL+1.600 x LL	1.00	1.00	1.20
204	UL:1.200 x DL+1.000 x LL	1.00	1.00	1.20
206	UL:1.200 x DL+0.800 x WLX	1.00	1.00	1.20
208	UL:1.200 x DL+0.800 x WLZ	1.00	1.00	1.20
214	UL:1.200 x DL+1.000 x LL+1.600 x WLX	1.00	1.00	1.20
218	UL:1.200 x DL+1.000 x LL+1.600 x WLZ	1.00	1.00	1.20
222	UL:1.200 x DL+1.000 x LL+1.000 x ELX	1.00	1.00	1.20
224	UL:1.200 x DL+1.000 x LL+1.000 x ELZ	1.00	1.00	1.20
226	UL:0.900 x DL+1.600 x WLX	1.00	1.00	0.90
228	UL:0.900 x DL+1.600 x WLZ	1.00	1.00	0.90
230	UL:0.900 x DL+1.000 x ELX	1.00	1.00	0.90
232	UL:0.900 x DL+1.000 x ELZ	1.00	1.00	0.90

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	189.646	-10.838	8.178	3.574	6.048
102	195.803	-10.484	8.197	3.577	6.010
105	194.264	-10.572	8.192	3.577	6.019
107	188.130	-10.322	7.250	3.210	4.567
109	203.286	-10.101	20.449	27.121	4.466
111	186.368	-7.896	8.167	3.544	2.353
113	197.766	-10.924	13.467	14.828	6.098
115	193.127	-10.185	7.496	3.303	4.908
119	204.494	-10.019	17.395	21.236	4.833
123	191.805	-8.366	8.183	3.554	3.248
127	200.354	-10.637	12.159	12.017	6.057
131	112.272	-5.987	3.979	1.780	2.148
133	127.428	-5.766	17.177	25.691	2.047
135	110.509	-3.561	4.895	2.114	-0.066
137	121.908	-6.589	10.196	13.398	3.679

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	265.504	-15.173	11.449	5.004	8.467

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
202	237.426	-12.438	9.843	4.294	7.196
204	233.732	-12.651	9.832	4.292	7.219
206	226.363	-12.593	9.071	3.998	6.072
208	238.487	-12.416	19.630	23.126	5.992
214	231.307	-11.826	8.347	3.710	4.850
218	255.556	-11.472	29.465	41.967	4.689
222	229.049	-8.448	9.816	4.249	1.941
224	245.332	-12.774	17.389	20.369	7.291
226	168.257	-8.929	5.875	2.634	3.073
228	192.505	-8.575	26.993	40.891	2.913
230	165.998	-5.551	7.344	3.173	0.165
232	182.282	-9.877	14.917	19.293	5.515

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.690m^2

Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 1.193m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 101

Width (W_2) = 1.300 m Governing Load Case : # 101

Depth (D_2) = 0.400 m Governing Load Case : # 218

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

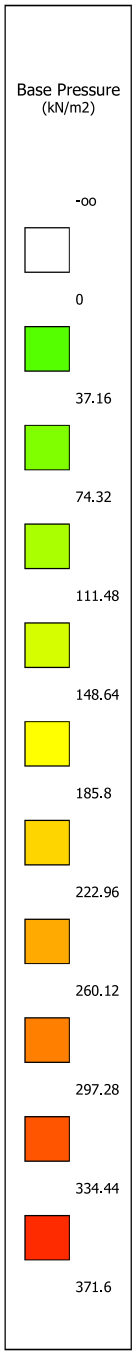
Gross Soil Bearing Capacity = 371.60 kN/m2

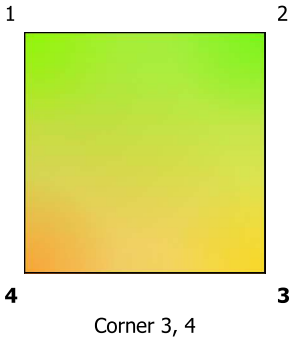
Soil Weight On Top Of Footing = 21.743 kN

Pressures at 4 Corners

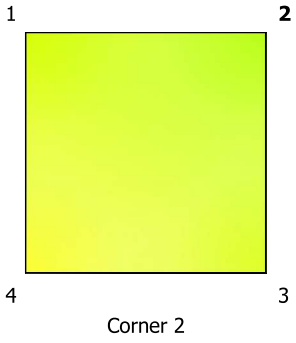
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

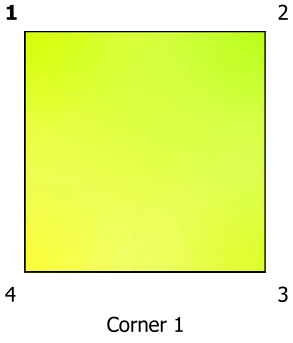




Corner 3, 4

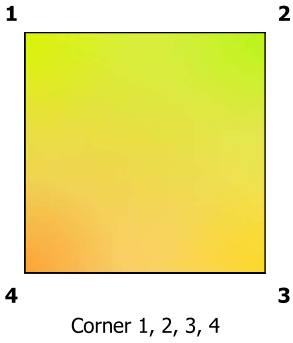


Corner 2



Corner 1

Governing Base Pressure Contour



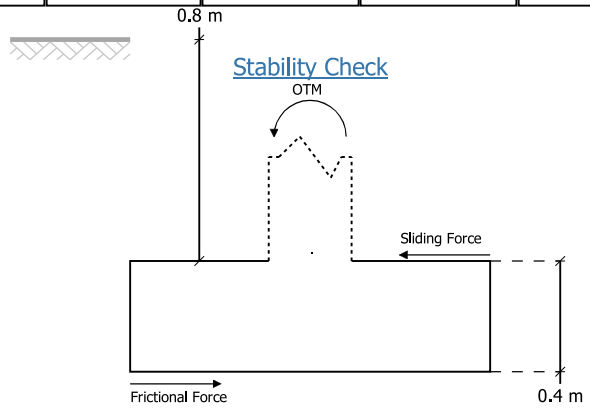
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
102	**147.4656**	91.7363	129.1846	184.9139	0.000
123	135.3235	**99.3046**	136.5955	172.6145	0.000
109	69.5801	23.1150	**215.9261**	262.3911	0.000
109	69.5801	23.1150	215.9261	**262.3911**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
102	**147.4656**	91.7363	129.1846	184.9139
123	135.3235	**99.3046**	136.5955	172.6145
109	69.5801	23.1150	**215.9261**	262.3911
109	69.5801	23.1150	215.9261	**262.3911**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	12.181	16.143	13.751	1.500	21.613	14.371	1.500
102	12.933	16.542	14.409	1.500	22.163	15.016	1.500
105	12.740	16.442	14.242	1.500	22.025	14.853	1.500
107	12.704	18.088	14.702	1.500	24.053	17.046	1.500
109	13.853	6.843	8.676	1.500	4.442	18.582	1.500
111	16.478	15.932	16.199	1.500	21.410	26.686	1.500
113	12.516	10.152	11.150	1.500	7.580	14.758	1.500
115	13.160	17.881	14.989	1.500	23.836	16.863	1.500
119	14.035	8.084	9.907	1.500	5.590	17.969	1.500
123	15.930	16.285	16.105	1.500	21.875	22.840	1.500
127	12.995	11.368	12.100	1.500	9.177	15.145	1.500
131	13.712	20.633	16.151	1.500	27.285	20.421	1.500
133	15.762	5.291	7.094	1.500	3.128	23.568	1.500
135	22.768	16.561	18.940	1.500	22.310	67.447	1.500
137	13.308	8.600	10.215	1.500	5.623	15.681	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : -10.838kN
Governing Restoring Force : 132.015kN
Minimum Sliding Ratio for the Critical Load Case : 12.181
Critical Load Case for Overturning about X-Direction : 133
Governing Overturning Moment : 32.562kNm
Governing Resisting Moment : 101.851kNm
Minimum Overturning Ratio for the Critical Load Case : 3.128

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 133
Governing Disturbing Force : 17.177kN
Governing Restoring Force : 90.884kN
Minimum Sliding Ratio for the Critical Load Case : 5.291
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : 10.383kNm
Governing Resisting Moment : 149.209kNm
Minimum Overturning Ratio for the Critical Load Case : 14.371

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

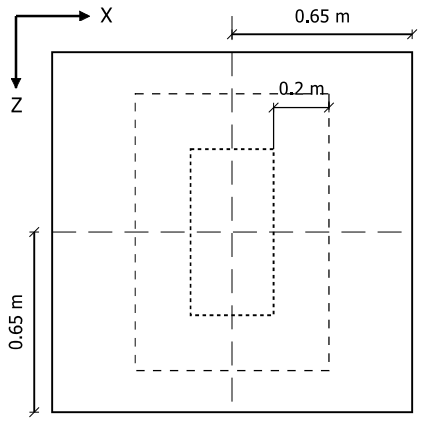
Critical Load Case for Sliding along Resultant Direction : 133
Governing Disturbing Force : 18.119kN
Governing Restoring Force : 128.530kN
Minimum Sliding Ratio for the Critical Load Case : 7.094

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m
Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m
For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c ≥ Punching Shear Force

Punching Shear Force, Vu = 204.029kN, Load Case # 201

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.656kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.802kN

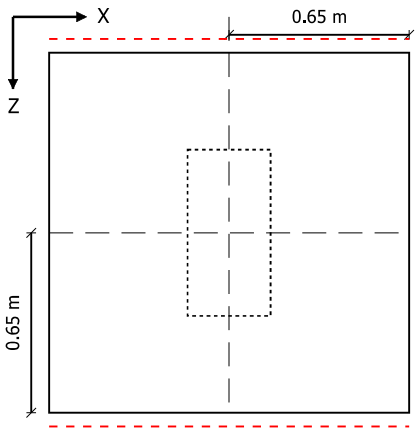
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.102kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.006m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

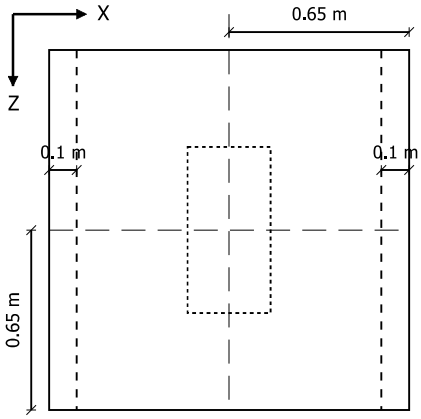
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 218 $V_{ux} = V_{ux}|_{x=D_x}$ = 2.530 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

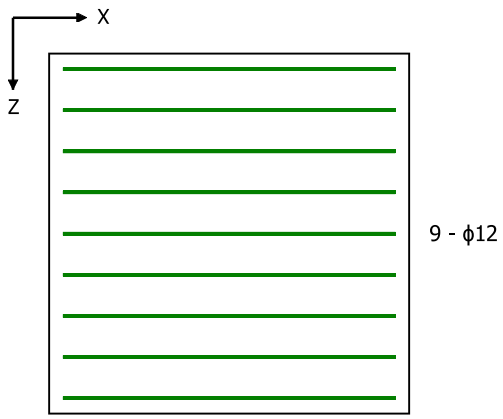
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 201 $V_{uz} = V_{uz}|_{z=D_z}$ = 45.324 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 201

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00177
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.500 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	35.435 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	39.372 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00061
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00053
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	918.298 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.450	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	9	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.869	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00228	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

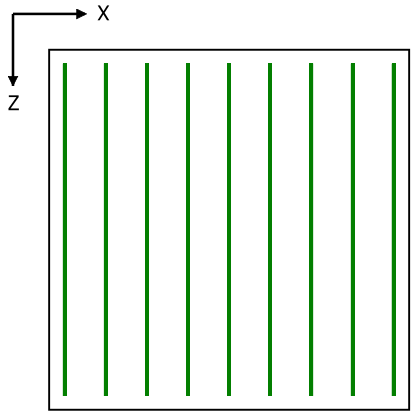
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.108

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 218

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl.7.12.2, ρ_{min} =		0.00174	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.950 m

Ultimate moment, $M_u|_{x=D_x}$ = 23.811 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 26.457 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00044

(Based on gross depth) ρ x d_{eff} / Depth = 0.00037

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 902.299 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.300 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

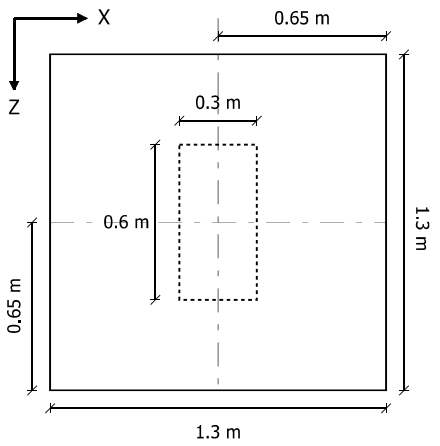
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

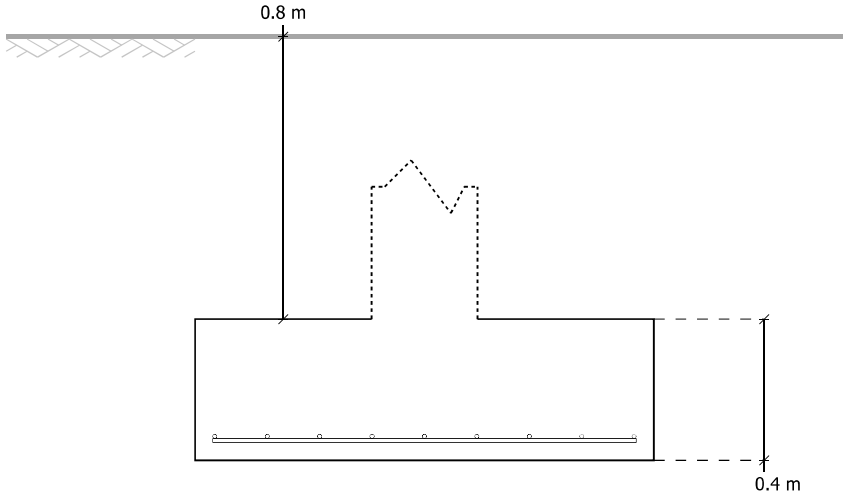
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.128

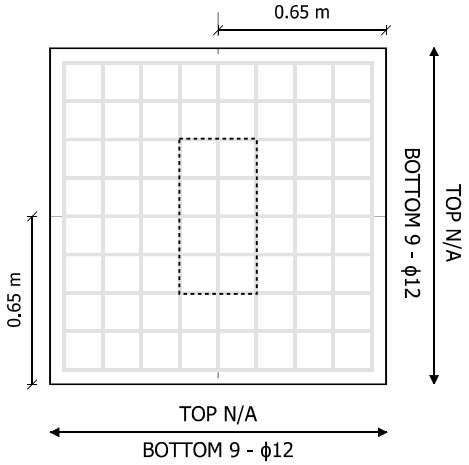
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

ATTACHMENT 3 COMMENT RESOLUTION LOG SHEET

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.:** 216
STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE: BECHTEL-WTRAN-001510, 02 March 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 02 March 2020	Comment by	EPC Response DATE: 10 March,2020	Status	Remarks
			General Comments (cc)				
MD-216-6000-EG-CV-CAL-1480-A01	6000	1	Please review if you require these beam for improving lateral stiffness of the building .	Mr.Klaus Grof	As per Table-10.12, SBC-301-2007, the Allowable Drift is H/100 i.,e 33mm where as attained values as per Analysis is 23mm (Refer Page-17 in MD-216-6000-EG-CV-CAL-1480-C01 for details) . Additionally, the HCS Slab diaphragm action with Concrete screed & extra joint reinforcement provided will further restrain the Building laterally. So the Secondary Beams can be neglected & will not be required.		